

Investor Day 2026

KLA

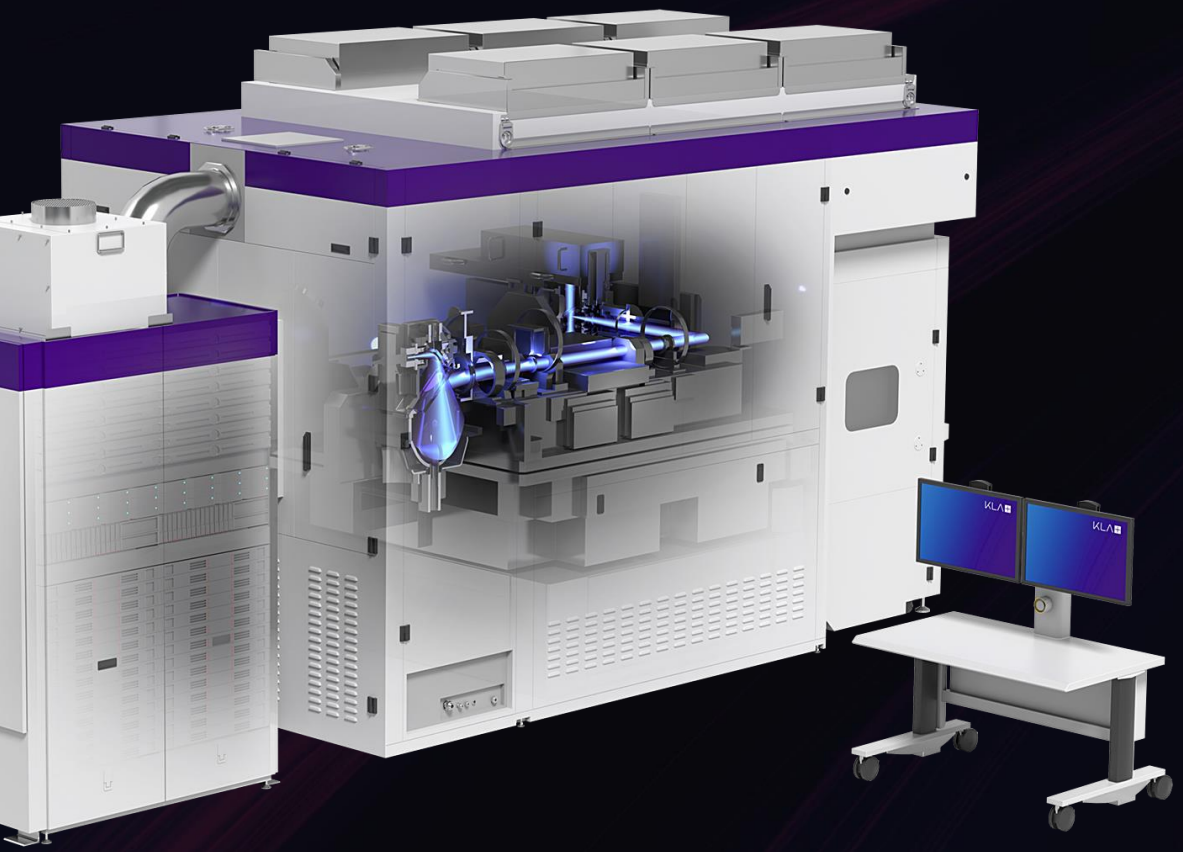
Process Control for the AI Era

Ahmad Khan

President, Semiconductor Products and Customers

March 12, 2026

Key Messages



1. **Inflections, share gains and portfolio expansion** will increase KLA's process control **intensity** from 7.4% (CY25) to ~9% (CY30)

2. Strong **customer collaboration** enables a broad, **differentiated product portfolio**, reinforcing KLA's competitive advantage

3. Unique, connected portfolio, driven by **innovations in hardware and AI**, provides customers with the best process control performance

CAGR (CY25 – CY30): Wafer Equipment 11% - 13% | KLA Systems 15% - 17%

KLA – Inspection & Metrology for 50 Years

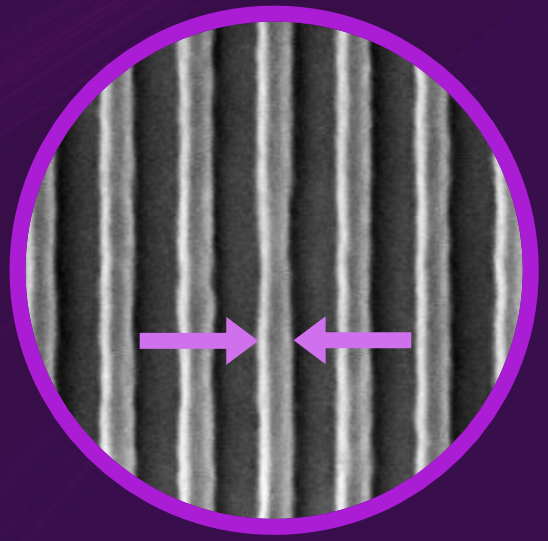
We Originally Thought it was About Finding Defects and Making Measurements – It's Not

Inspection
Find Critical Defects



You can't fix
what you can't find

Metrology
Measure Critical Parameters



You can't control
what you can't measure

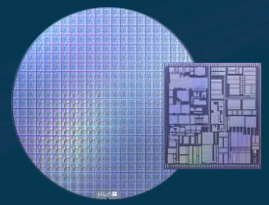
Learning faster is the only sustainable competitive advantage in semiconductor manufacturing

KLA Enables Device Yield and Device Performance

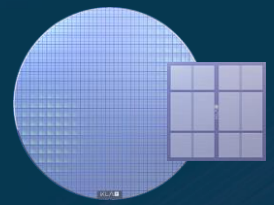
KLA Supports Markets Across the Semiconductor Ecosystem

Our Comprehensive Solutions and Expertise Play a Crucial Role in Enabling AI

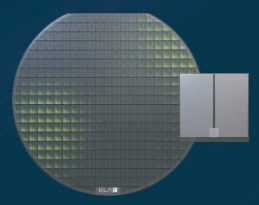
Chips



Logic | Foundry



Memory

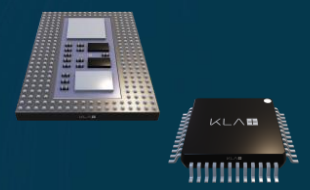


Specialty | Legacy

Wafer-Level Packaging



Components



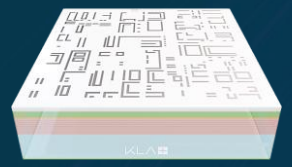
Printed Circuit Boards



Wafers



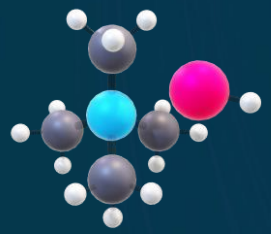
Reticles



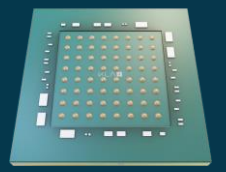
Semiconductor OEMs



Chemicals | Materials

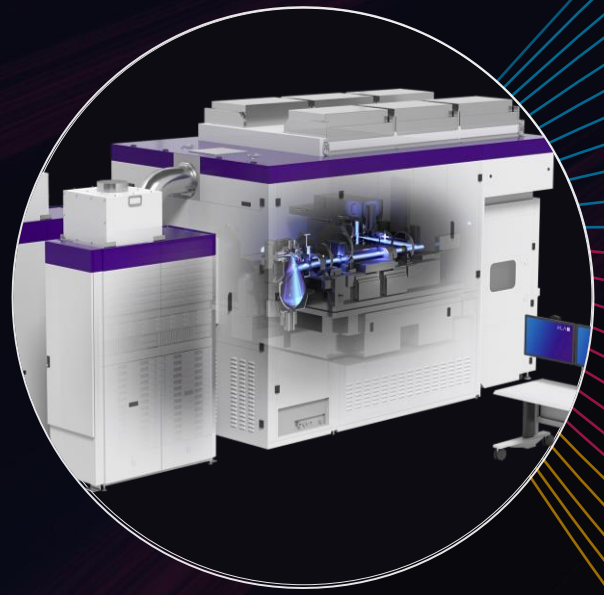


IC Substrates



KLA's Indispensable Process Control Portfolio Provides Solutions for Manufacturing Success

Systems and Co-Intelligence



technology innovation and data connections find variations and defects

Inspection

- BBP Patterned Wafer Inspection
- LS Patterned Wafer Inspection
- Macro Patterned Wafer Inspection
- All-Surface Patterned Wafer Inspection
- Unpatterned Wafer Inspection
- e-Beam Inspection and Review
- Reticle Inspection (Mask Shop)
- Reticle Inspection (IC Fab), Print Check
- Mask Blank Inspection
- Wafer-Level Packaging Inspection
- Packaged IC Component Inspection
- IC Substrate Inspection
- Printed Circuit Board (PCB) Inspection

Metrology

- Overlay Metrology
- Critical Dimension, 3D Shape Metrology
- Film Thickness, Roughness, Composition Metrology
- Patterned / Unpatterned Wafer Geometry
- Resistivity Metrology
- Implant / Anneal Metrology
- Electrical / Magnetic Property Metrology
- Reticle Registration Metrology
- Panel Warpage, Bow Metrology
- In Situ Process Management

Data Analytics

- Defect Analytic Software
- Litho | Etch Simulation Software
- Patterning Analytic Software
- Reticle Analytic Software
- Pattern-Centric Software
- CAM-Based Data Analytics Software



The KLA Operating Model Provides a Strategic Engine for Product Differentiation and Competitive Advantage

Strategy: Provide leading technologies, differentiated solutions and superior customer experiences by leveraging our operating model



Collaboration

KLA builds strong customer partnerships, producing solutions that address the industry's critical technology inflections and economic challenges

Innovation

KLA's world-class point products and co-intelligent solutions enabled by AI deliver faster ramp and higher yield, ensuring customer profitability

Execution

KLA's strong presence in R&D and HVM for logic, memory and advanced packaging drives process control intensity, while our product differentiation secures clear market leadership

Systems Market-Leading Performance

Our Ability to Uniquely Solve Our Customers' Critical Challenges Drives Success

$9.8B
Systems Revenue[1]
CY25

$9.0B
Semi PC Revenue[1]
CY25

19%
Semi PC Revenue CAGR
CY19 – CY25

62.8%
KLA Gross Margin[1]
CY25

$1.44B
R&D[1]
CY25 R&D spend for KLA

>65%
PhD/Master's Among Professional Roles

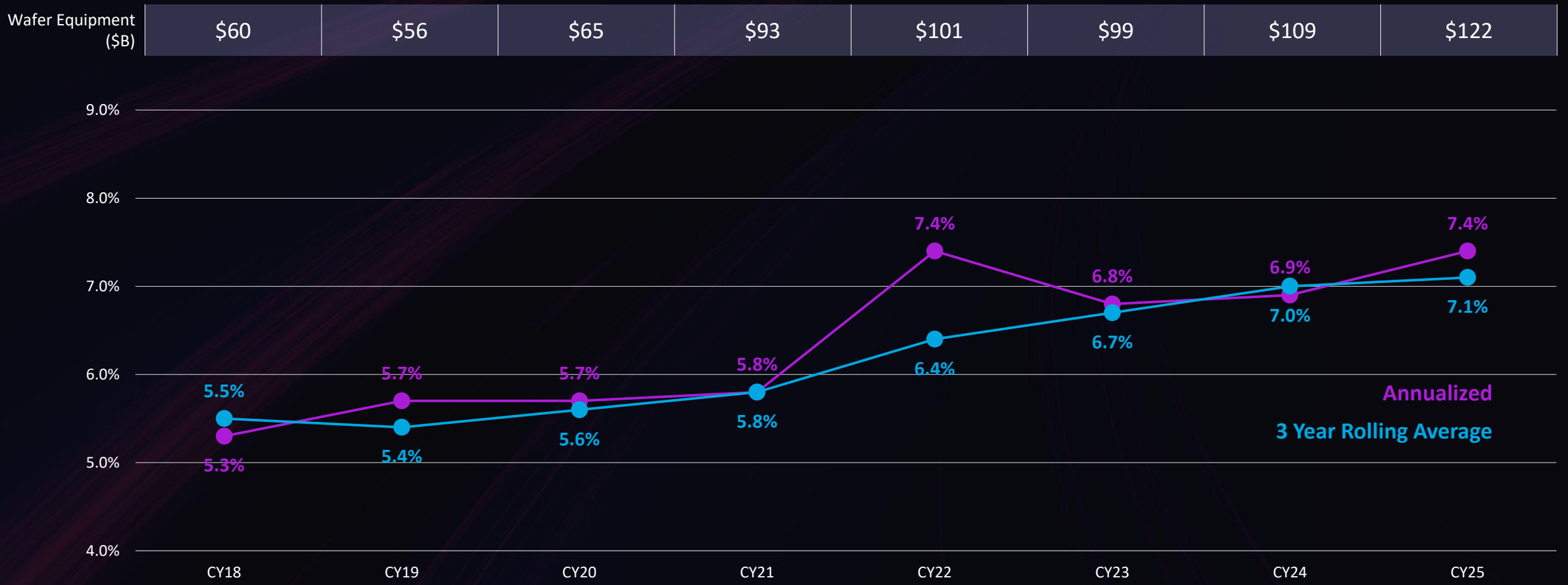
56.5%
Process Control Market Share[2]
CY24

7.4%
Process Control Intensity
CY25

1: KLA filings – non-GAAP, reconciled GAAP figures in appendix ; 2: Gartner April 2025

KLA Semi PC Share of Wafer Equipment

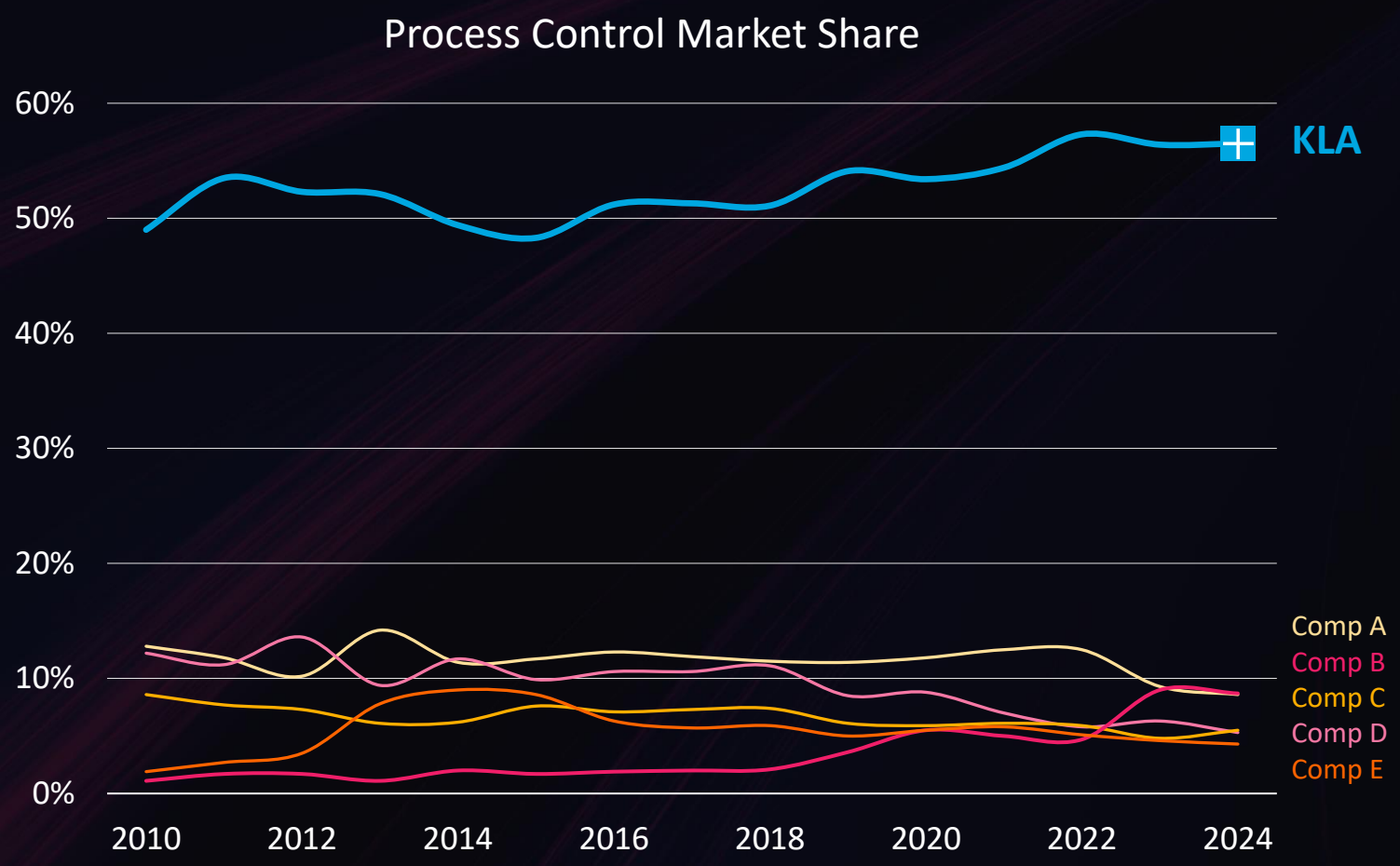
Multiple Device Segments in R&D and HVM Drive New +7% Baseline



Wafer Equipment includes WFE (Wafer Front End) and WLP (Wafer-Level Packaging)

KLA Continues to Decisively Lead the Industry

#1 Market Position is Driven by Technology Differentiation and Strength in Execution



56.5%

KLA Share in 2024

- Up 2% from 2021
- 6.5x nearest competitor
- Gains in optical, e-beam, advanced packaging, reticle inspection

Source: Gartner April 2025

Multiple Process Control Segments Drive Intensity and Share

Higher Cost of Failure in Today's AI-Driven Semiconductor Landscape

Reticle Inspection

- 100% defect capture is mandatory
- Multiple inspection products are required to capture 100% of defect types

Unpatterned Inspection

- More process steps, process tools and qualification are driving baseline defectivity importance

Optical Inspection

- More layers
- More defect types
- Higher sampling
- New applications

e-Beam Inspection

- 3D architectures introduce buried and electrical defects in HVM
- Co-intelligent solutions lead to faster time to results for optical systems in HVM

Metrology

- Critical films metrology for Vt control optimizes device power and performance

Advanced Packaging

- Packaging delivers final revenue
- Packaging yield is a sum of all parts: front-end, interposer and stacking

Data Analytics

- Fab-wide process control enables faster ramps and higher yield

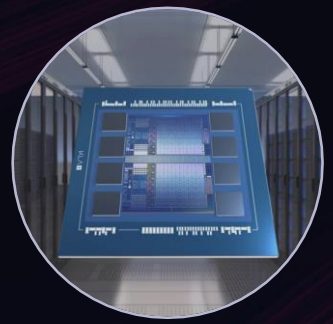


Process Control Applications Engineering Experts Drive Customer Collaborations

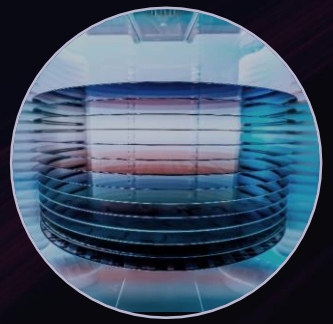
The KLA Operating Model Drives Semi PC Success



Customer Collaboration

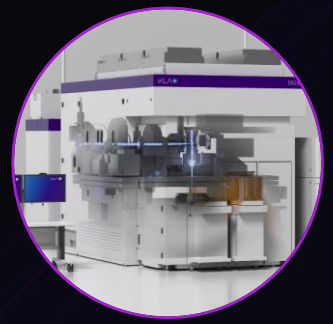


Technology Inflections



High-Volume Manufacturing

Product Innovation



Point Product Differentiation



Co-Intelligent Solutions

Execution

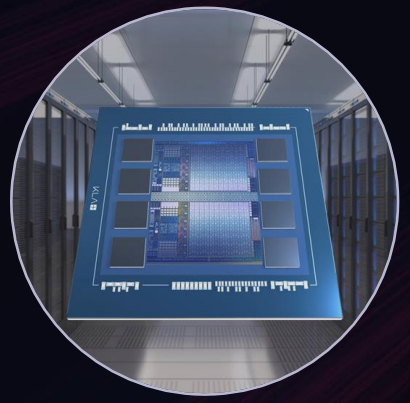


2030 Target Model

The KLA Operating Model Drives Semi PC Success

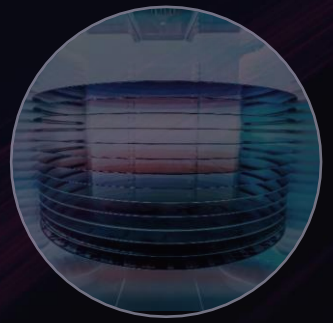


Customer Collaboration



Technology Inflections

AI-Based Inflections and Chip Complexity Drive Process Control Intensity and Share

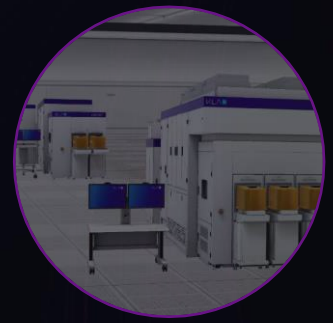


High-Volume Manufacturing

Product Innovation

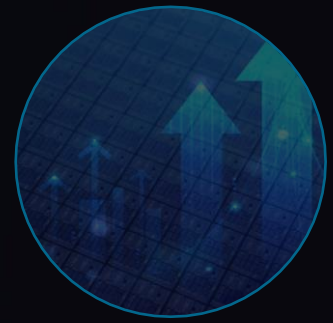


Point Product Differentiation



Co-Intelligent Solutions

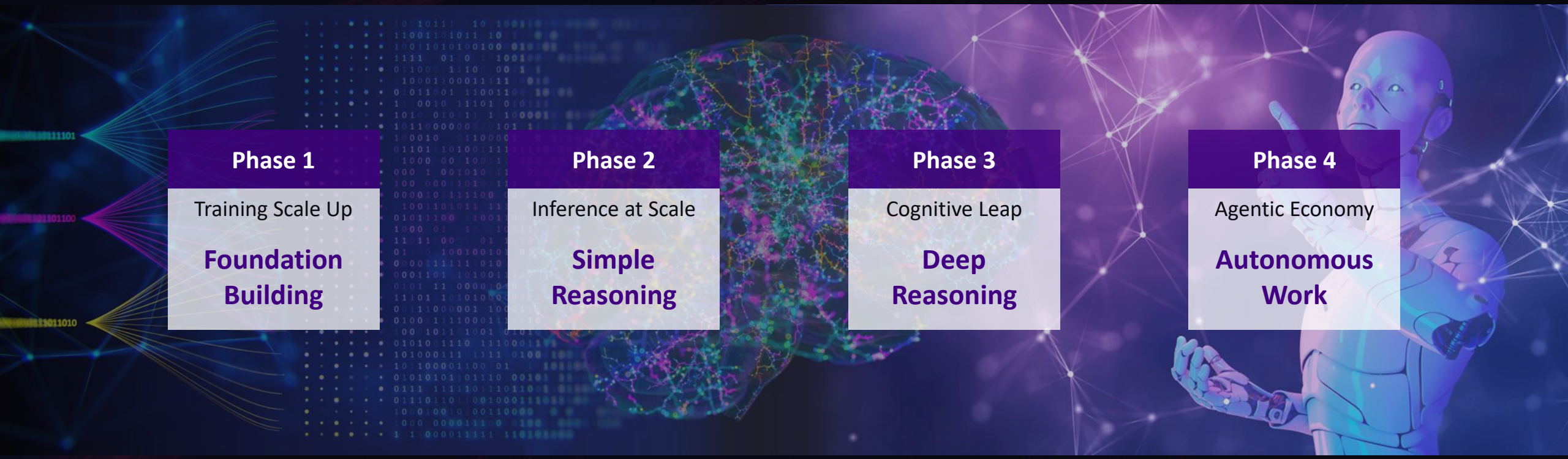
Execution



2030 Target Model

Evolution of AI

AI Growth Will Drive Significant Demand for Compute Resources

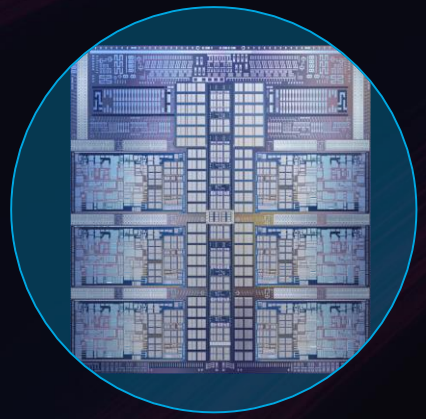


Semiconductor Enablers of AI Evolution

AI Drives Growth in All Device Segments

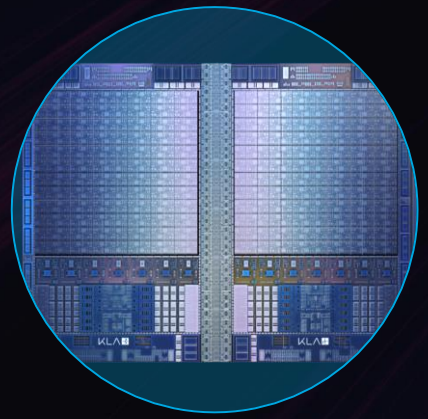
CPU

Orchestration



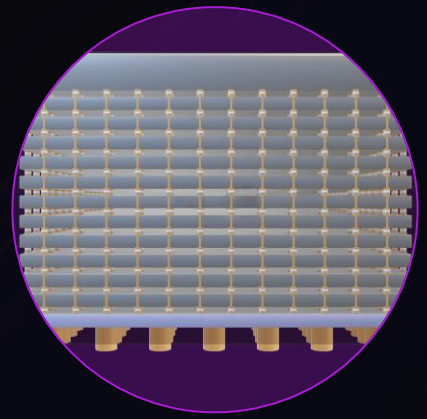
GPU

AI Compute



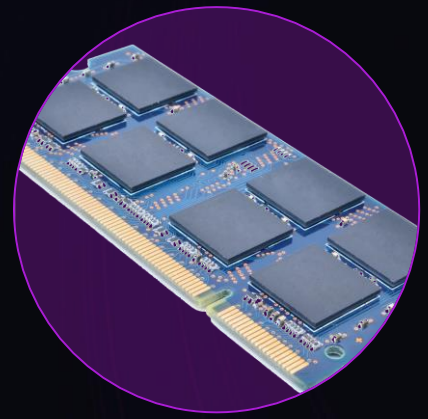
HBM

Weights
Key-Value (KV) Cache



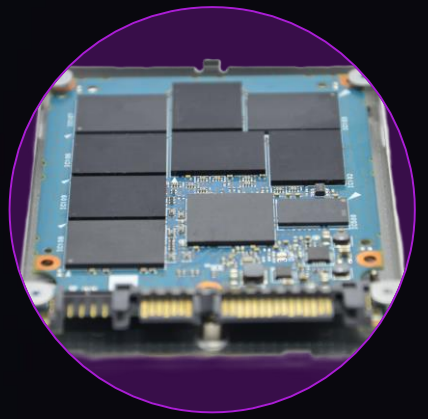
DRAM

Conversations



Flash

Conversation History



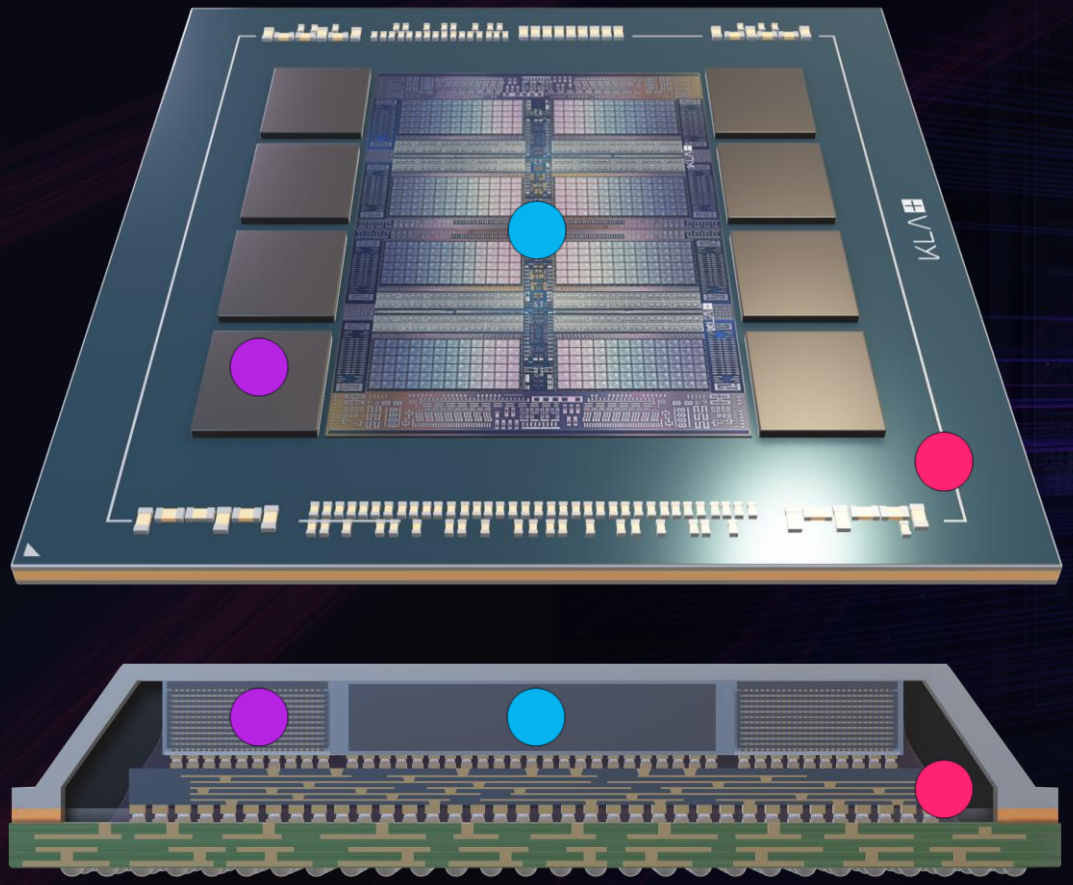
Logic

Memory

AI Accelerates Semiconductor Innovation

Multiple, Leading-Edge Chips Are Assembled to Form a Complete System

AI Accelerator for Server

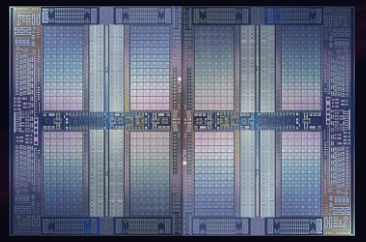


Leading-edge logic GPUs

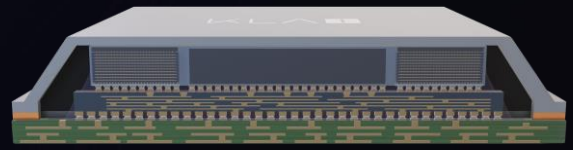
Multiple HBM modules, 8 to 12 chip stacks

Large, complex package using 2.5D/3D heterogeneous integration and ~100 chips

AI-Driven Inflections Accelerate Challenges for Our Customers







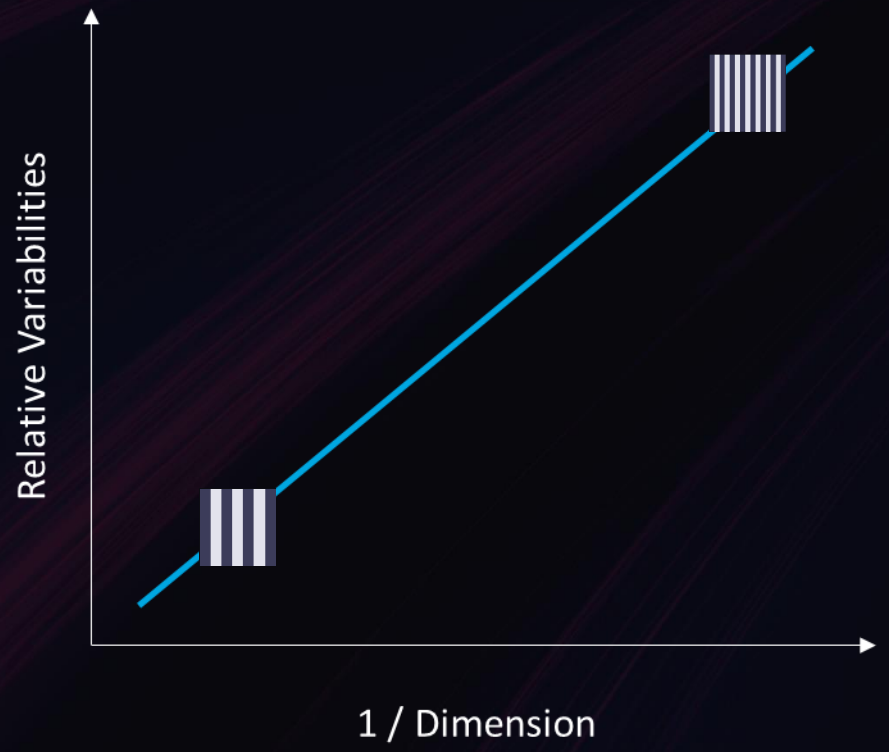
Advanced Logic	Memory	Advanced Packaging
▪ Smaller pattern dimensions	▪ More logic, less pattern redundancy	▪ Chip stacking
▪ Larger chip size	▪ Larger chip size	▪ Rising package value
▪ 3D transistor architectures	▪ EUV	▪ Interconnect scaling

Critical Defect Monitoring and Increased Variability Control are Required for Advanced Logic Manufacturing

Logic Inflections

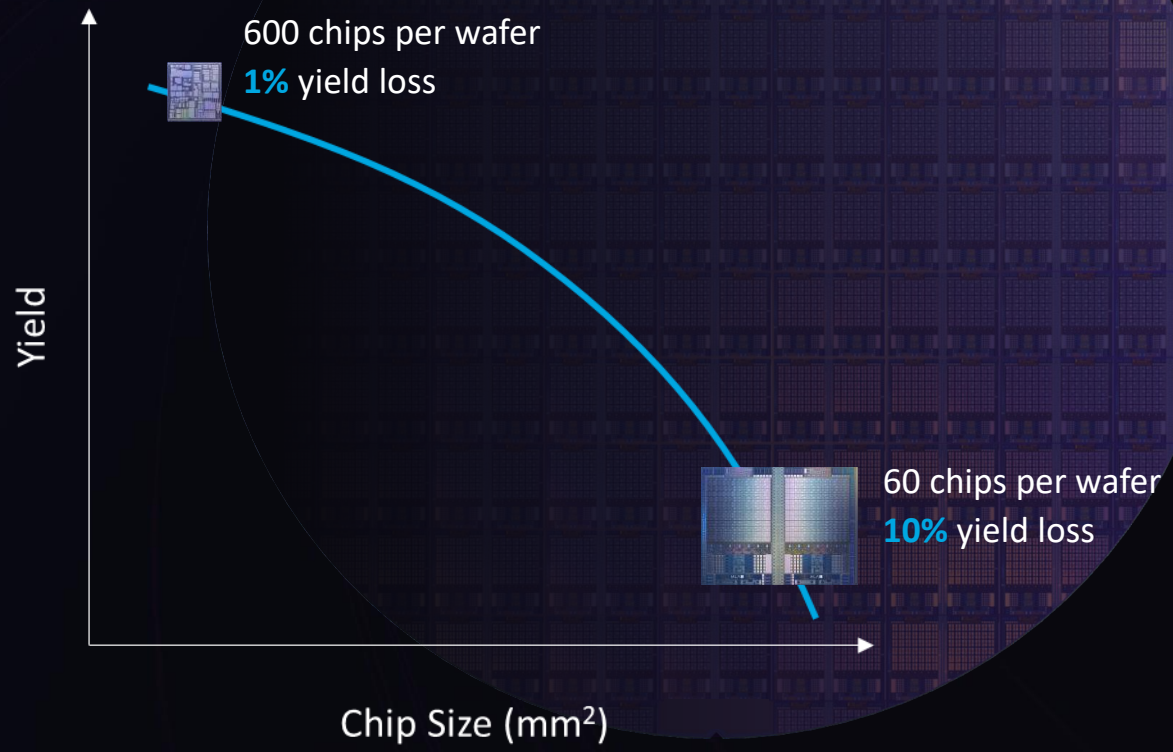
Smaller Pattern Dimensions

- Variability is inversely proportional to printed feature size



Larger Chip Sizes

- Steep yield erosion on larger chip sizes



Complex Transistor Architectures Increase Demand for Defect and Variability Control

Logic Inflections

Inspection

find critical defects

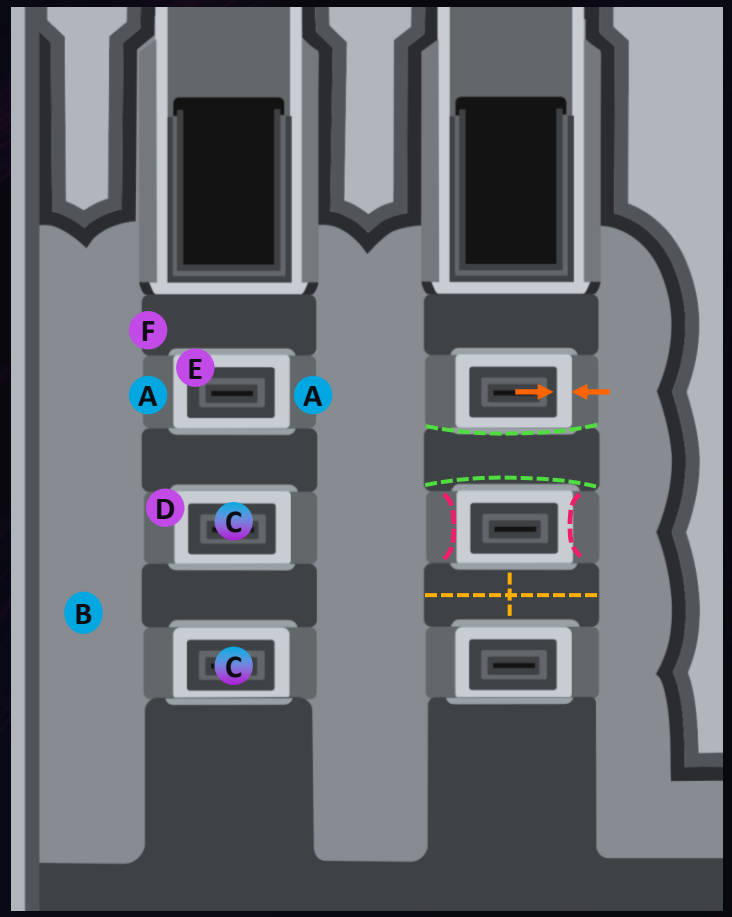
Increase Yield

Buried defectivity

a. Spacer under etch residue
b. Source / Drain missing epi
c. SiGe residue
d. HK / work function metal filled epi
e. HKDP gate extrusion
f. Bright voltage contrast at epi layer

BBP optical defect inspection
e-beam defect inspection

3D Transistor Inflections



Metrology

measure critical parameters

Increase Performance

Work function metal and HKMG thickness

Nanosheet release integrity

Inner spacer and recess depth

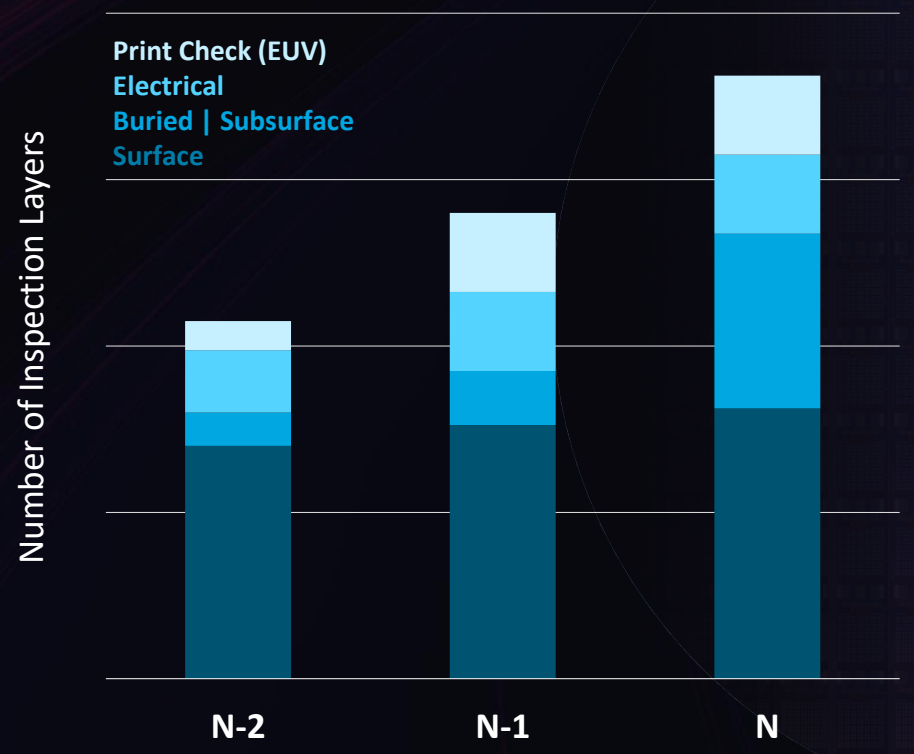
Nanosheet CD and thickness

KLA's Products Deliver Solutions for Critical Logic Challenges

Collaboration

- Comprehensive BBP and e-beam inspection portfolio provides capture of all critical defect types at optimized cost of ownership
- Multi-system solution for Ångstrom-level critical device measurements enables variability and Vt control
- Multi-system reticle requalification strategy with reticle inspection and wafer print qualification reduces high-volume manufacturing yield risk

Patterned Wafer Inspection



KLA internal analysis

30%

Increase in Inspection Layers with Every Node

~$1B

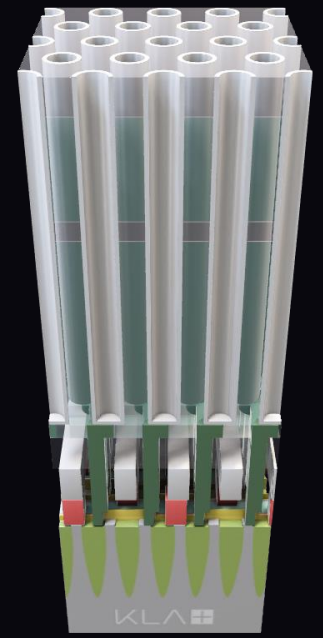
KLA Incremental Revenue vs. N-2[1]

1: For equivalent wafer starts

DRAM and HBM Memory Chip Designs Drive High Yield and Reliability Requirements

Memory Inflections

	Higher Performing Logic Circuitry	EUV Lithography	Large Chips	Less Circuit Redundancy
Technology Driver	Better logic performance is required to support smaller capacitors	Patterning advancement for smaller design rules	More logic, more DRAM transistors and more TSVs produce higher bandwidth	Percentage of cell area decreases
Implications for Process Control	▪ More logic requires improved capture of defects ▪ Higher variability control	▪ Smaller defects print ▪ New process control applications (e.g., reticle qualification)	▪ One killer defect has a higher impact on yield	▪ Higher yield impact of defects



Today's Server Memory Requirements are Much More Stringent than Consumer Memory

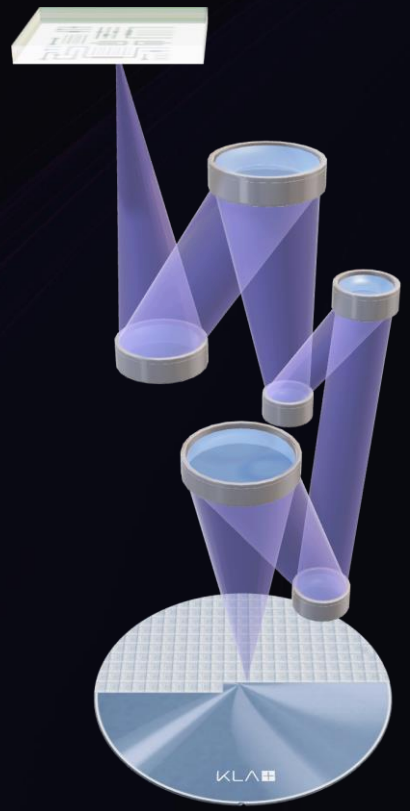
DRAM and HBM Memory Chip Designs Drive High Yield and Reliability Requirements

Memory Production Trends

More process control is required to mitigate rising variability and the increased severity of defects associated with:

- Smaller pattern features (EUV)
- DRAM chips with more logic content and less redundancy
- Larger chip sizes

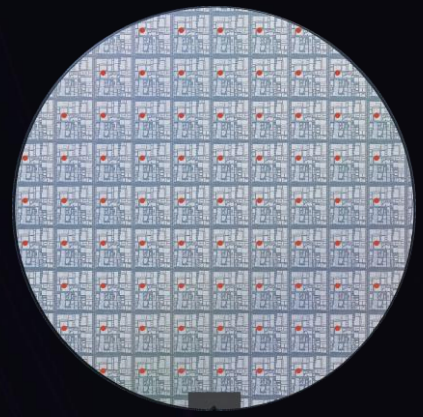
EUV Adoption: DRAM



EUV Reticle Requal



New Wafer Print Check Applications



DRAM and HBM Memory Chip Designs Drive High Yield and Reliability Requirements

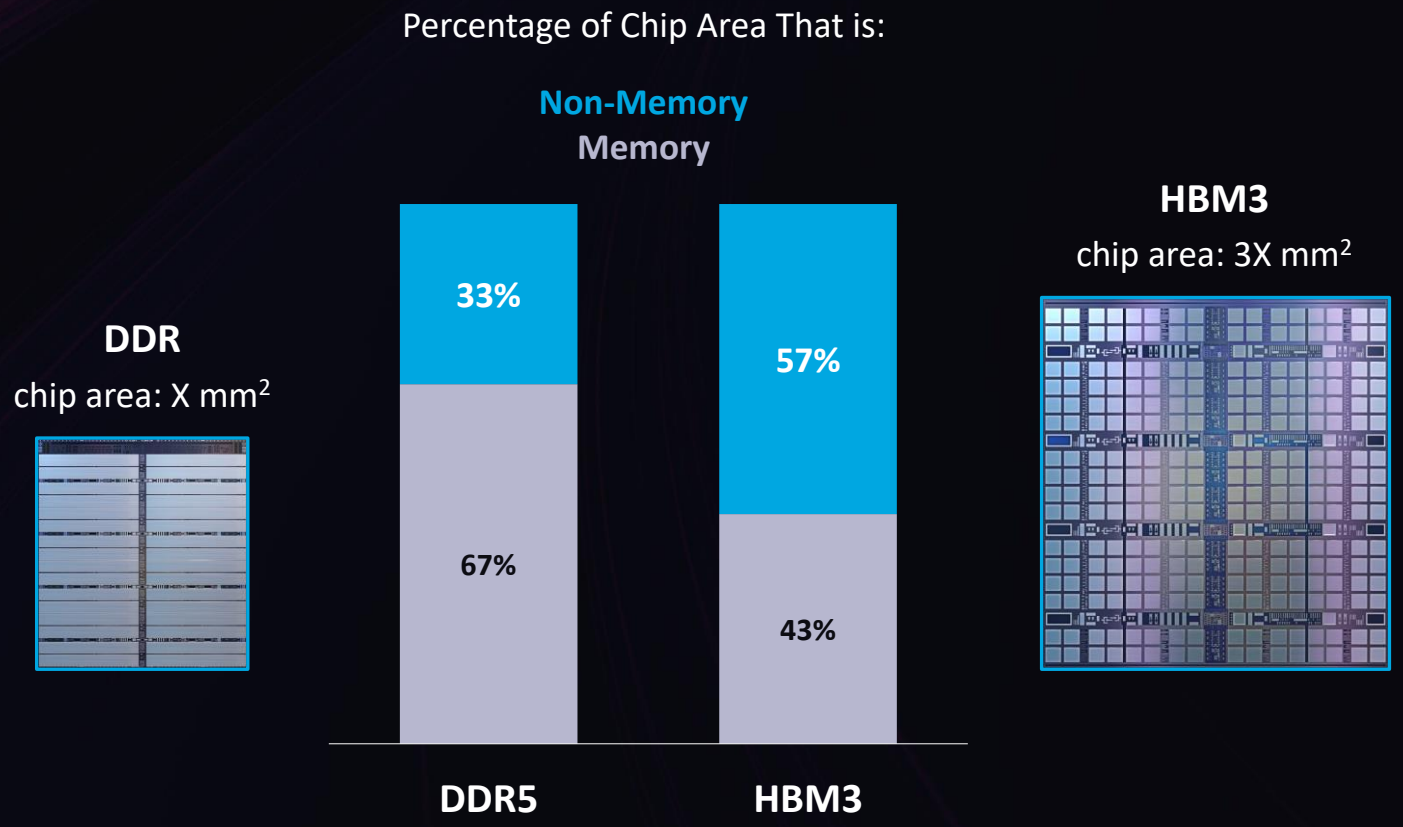
Memory Inflections

Memory Production Trends

More process control is required to mitigate rising variability and the increased severity of defects associated with:

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- DRAM chips with more logic content and less redundancy
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Less Redundancy, Larger Chip Size: HBM



Sources: TechInsights

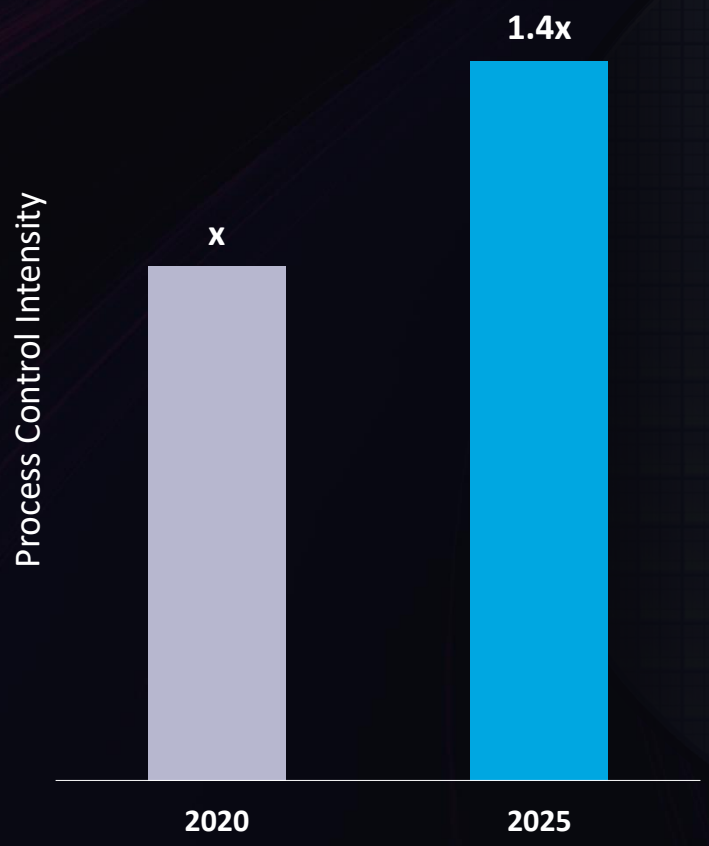
KLA's Products Deliver Solutions for Critical Memory Challenges

Collaboration

- BBP optical and e-beam inspection synergy enables fast production ramp
- New Gen4 and Gen5 BBP optical inspection platforms provide sensitivity to critical defects at optimal cost of ownership
- Multi-system reticle requalification strategy with reticle inspection and wafer print qualification reduces high-volume manufacturing yield risk
- Process control expertise leveraged to optimize high-volume manufacturing sampling strategies to address the yield risk associated with reduced pattern redundancy

Increasing DRAM Intensity

(% of Wafer Equipment)



$450M

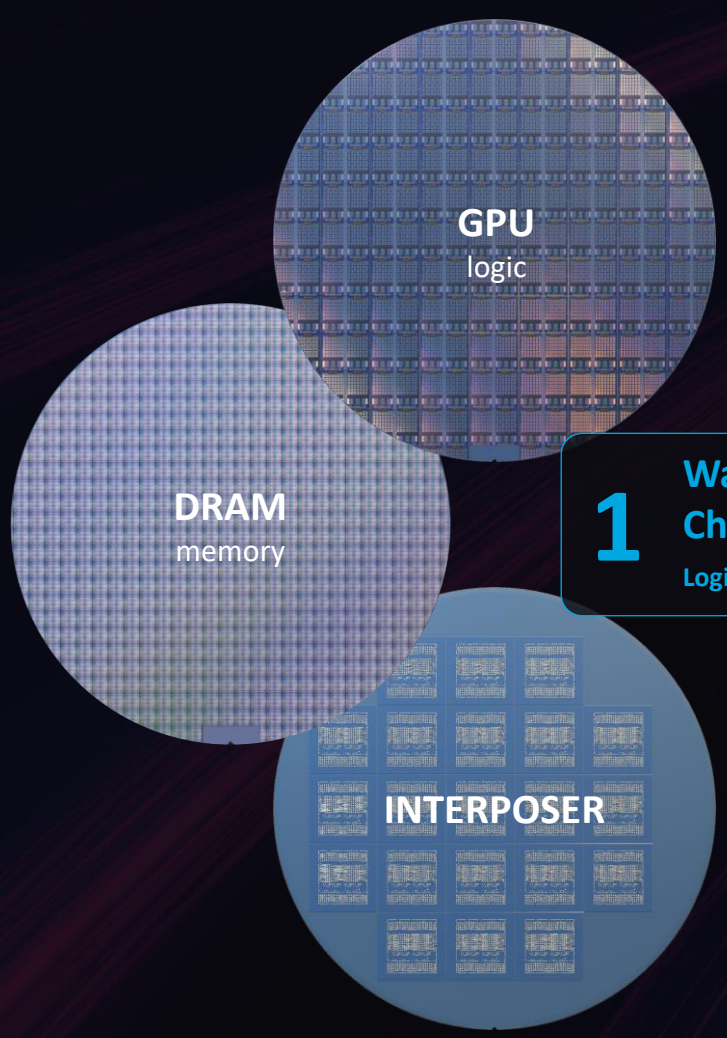
KLA Incremental Intensity Gain in CY25

From CY20 to CY25:

- **DRAM Wafer Equipment ~2x**
- **KLA DRAM Revenue ~3x**

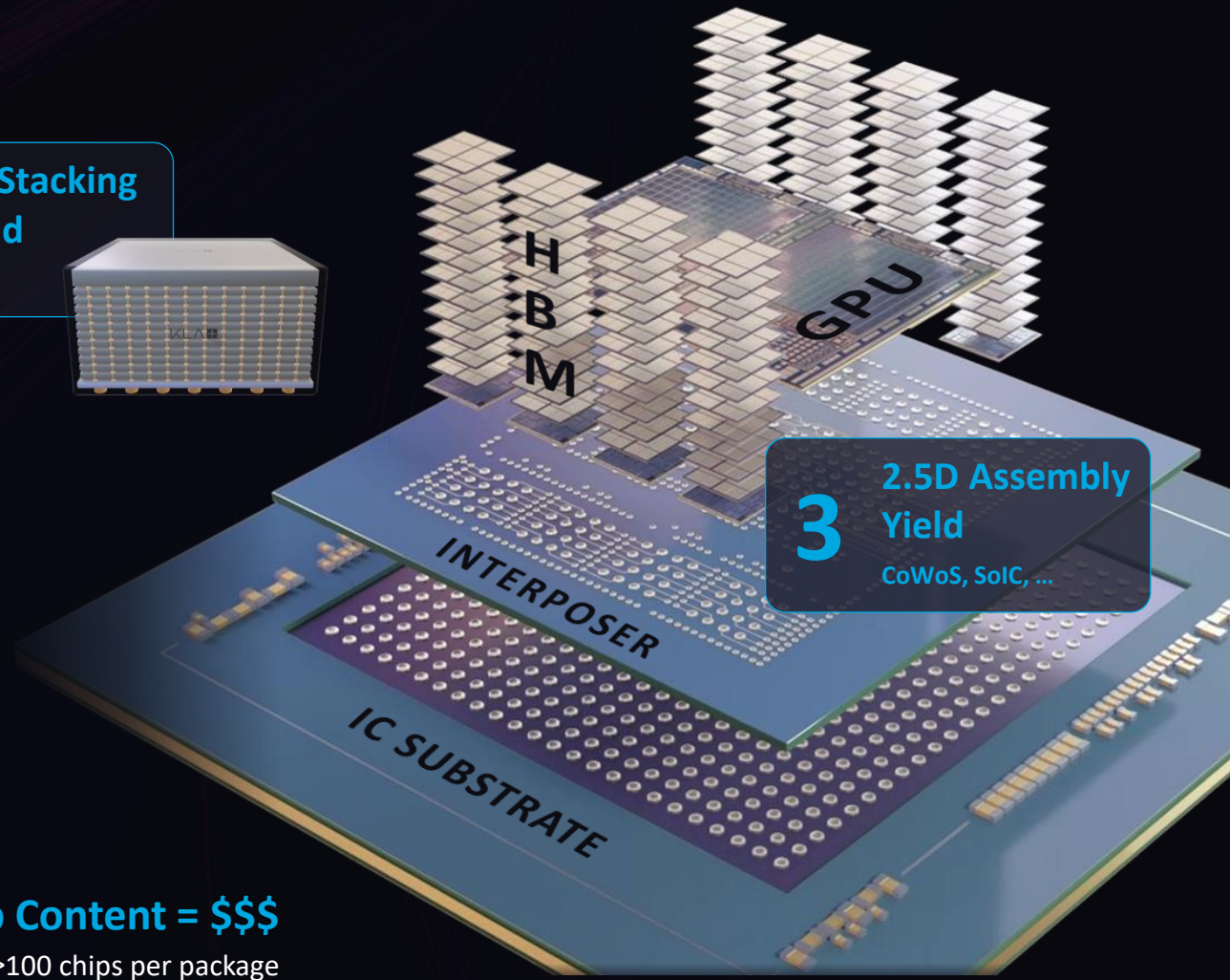
Advanced Packaging Compounds Integration Complexity and Potential Yield Loss

Packaging Inflections



1 Wafer-Level Chip Yield
Logic, Memory, Interposer

2 3D Stacking Yield
HBM



3 2.5D Assembly Yield
CoWoS, SoIC, …

High Chip Content = $$$
>100 chips per package

KLA's Products Deliver Solutions for Advanced Packaging Ramp

Packaging Inflections

2008



ICOS™



CIRCL™-AP

2015

- Technology and experience from front end process control systems
- Wafer handling innovations
- Acquisitions

Patterned Inspection
wafer, frame



Kronos™



Puma™



Voyager®



29xx

High Throughput ← → High Sensitivity



Surfscan®



CIRCL™-AP



eDR7380™



Micro-SR™



Lumina™



ICOS™ F260



ICOS™ T3|T7|T9



Archer™



PWG™



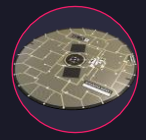
Aleris®



SpectraShape™



microRSP®



SensArray®



Klarity®



5D Analyzer®



QualiSurf®



Quali-Fill® Libra®



OmniMap®



Zeta™-5xx/6xx



Corus™



PROLITH™



SPTS Osprey®



SPTS Omega®



SPTS Mosaic™



SPTS Sigma®



Orbotech Precise™

Continued platform evolution and innovation to support future inflections

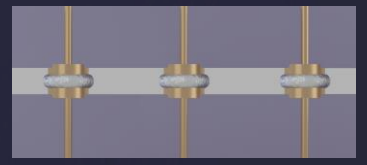
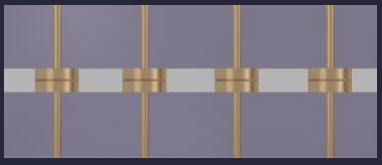


Interconnect Scaling in Advanced Packaging Drives Process Control Intensity

Packaging Inflections

Interconnect Scaling

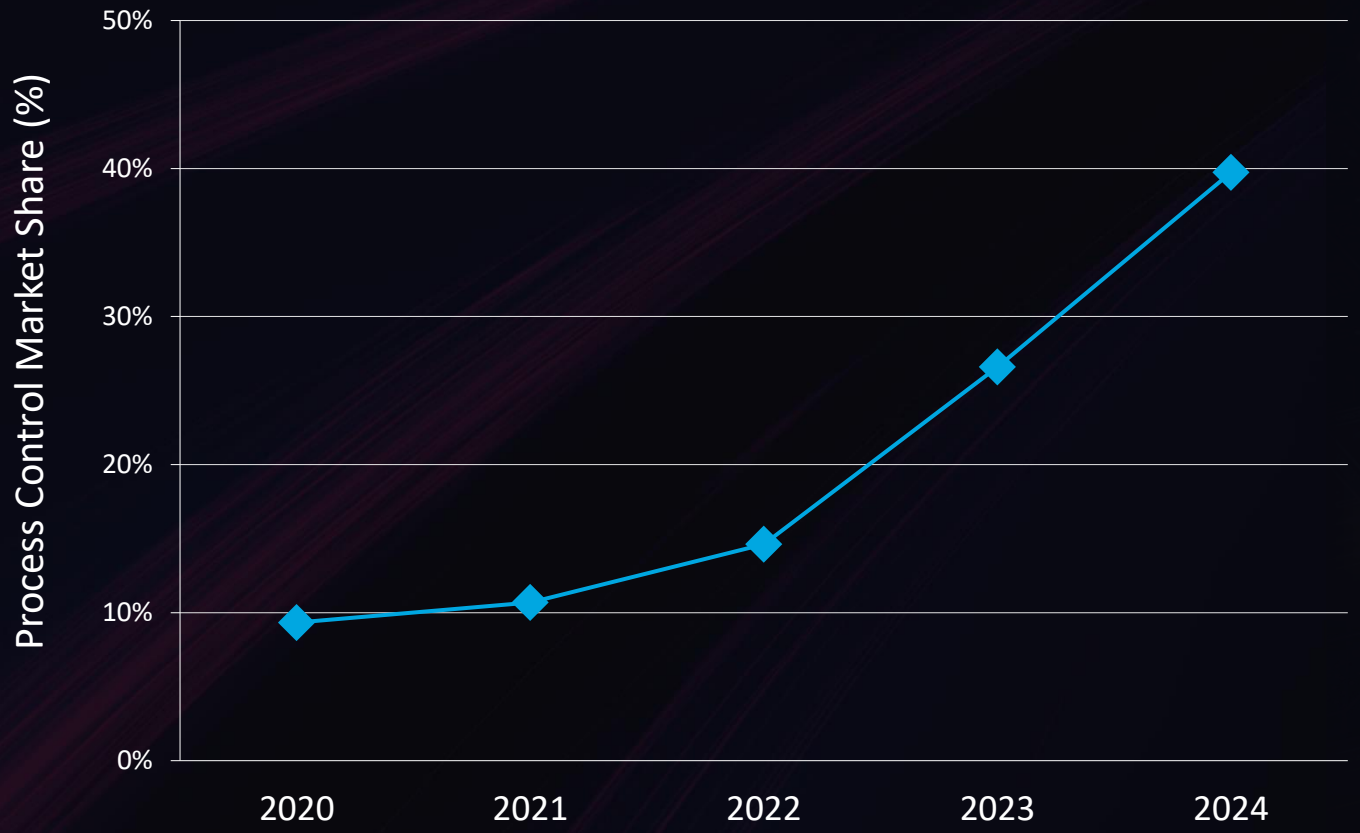
- Smaller 2D and 3D pattern dimensions and pitch require more sensitive inspection and more accurate metrology

	Microbump	Hybrid Bond
		
Bump Pitch (µm)	20 – 50	<10
Bump Height (µm)	10 – 25	-0.010* – 0
Diameter (µm)	10 – 25	<5
Defect Sensitivity (µm)	1 – 5	0.05 – 0.5
Overlay / Alignment Error (µm)	1 – 2	0.05 – 0.25
Metrology Accuracy (µm)	1	0.001*

*Cu recess

KLA's Products Deliver Solutions for Advanced Packaging Ramp

Close Customer Partnerships Resulted in ~4x Increase in Market Share in Four Years



~$950M

KLA Advanced Packaging Revenue in 2025

#1 Market Share

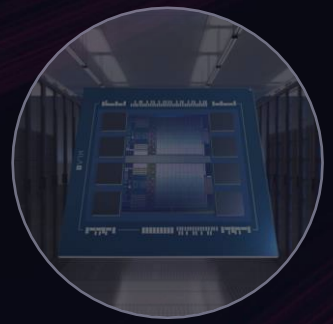
in Process Control in 2025

Source: TechInsights April 2025

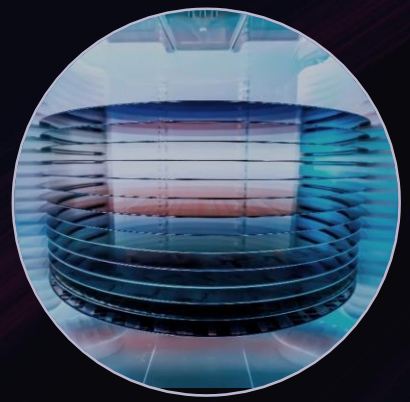
The KLA Operating Model Drives Semi PC Success



Customer Collaboration



Technology Inflections



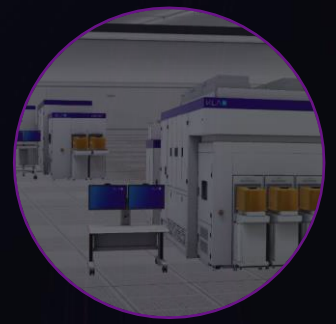
High-Volume Manufacturing

Production Variability and Increased Design Starts Require More Intensive Process Control

Product Innovation

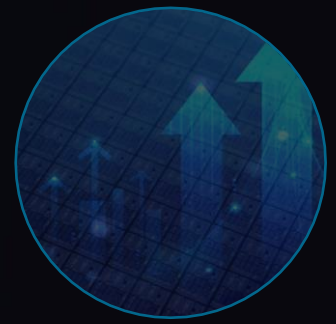


Point Product Differentiation



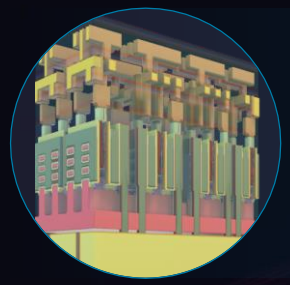
Co-Intelligent Solutions

Execution

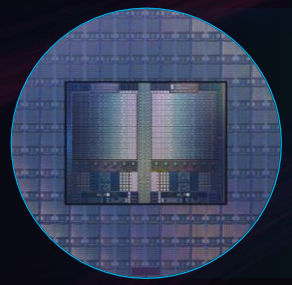


2030 Target Model

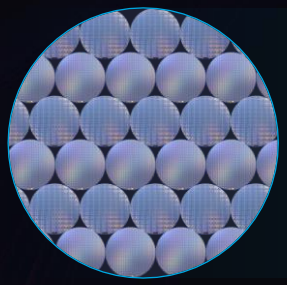
Relentless Challenges Faced by a High-Volume Manufacturing Fab



Fabricate complex device structures with nanoscale precision

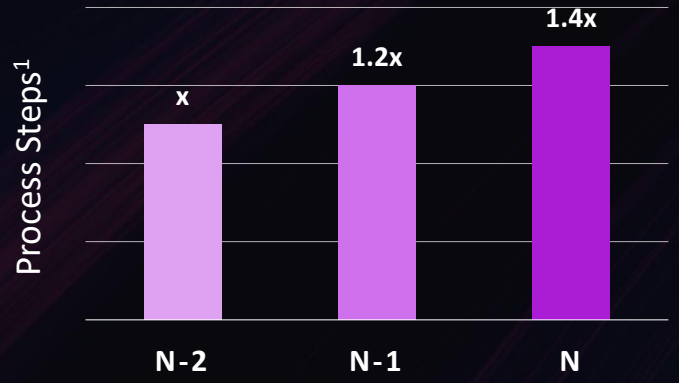


Control 100s of process tools that execute >1,000 process steps to produce functional chips

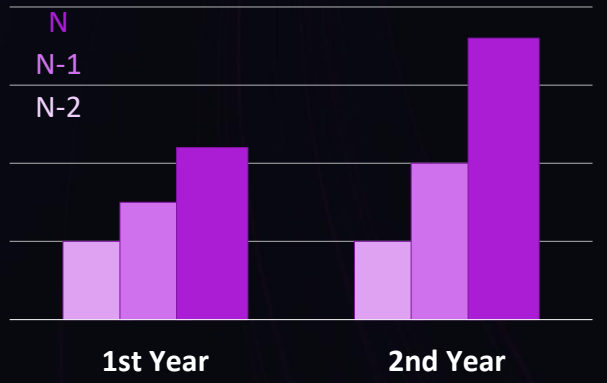


Maintain yield and quality for production of multiple designs at 100,000 WSPM

feature size ↓ | 3D architectures ↑ | process steps ↑ | design mix ↑ | chip size ↑ | inline control specs ↓ = process variability ↑



Longer and more complex process flows as design nodes advance



Broader design mix as nodes ramp[2]

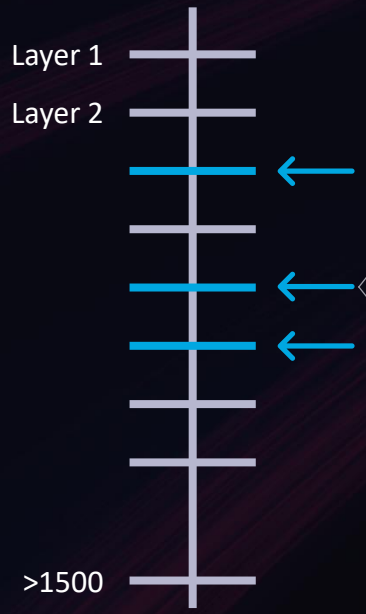
Process Control Is Critical: 1% Incremental Yield Delivers >$150M Incremental Annual Revenue for Leading-Edge

1: Adapted from IC Knowledge Strategic Cost Model; 2: Adapted from: TechInsights

Increased Sampling is Needed to Find and Monitor Yield Challenges

More Sampling Leads to Higher Adoption of Process Control

What Process Layers?



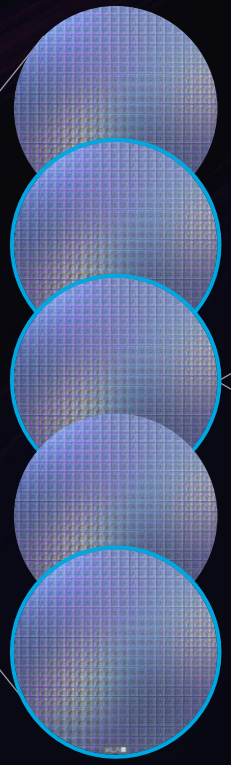
Inflections

EUV, GAA, HBM, …

% of Lots?



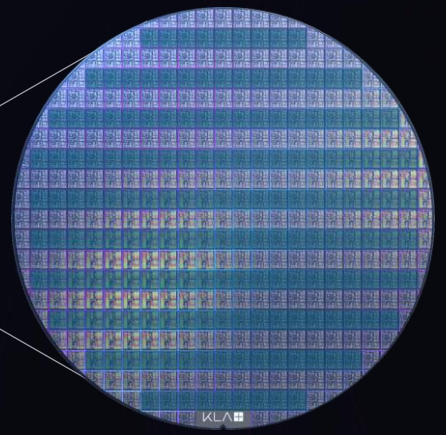
Wafers per Lot?



HVM Variability

Voyager®, Surfscan®, …

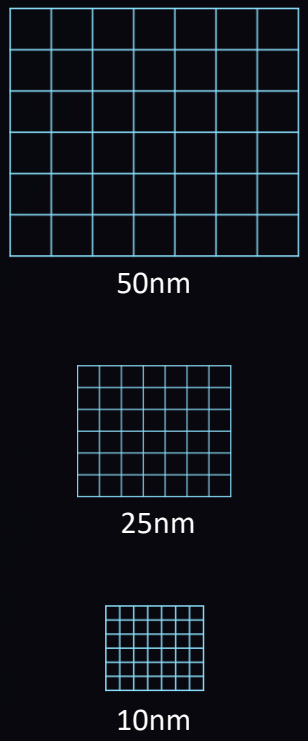
% of Wafer?



HVM Variability

Metrology Sites per Wafer

Sensitivity?



Smaller Pixels, Multi-Pass

Multi-Tool, …

KLA's Products Deliver Solutions for Leading-Edge High-Volume Manufacturing

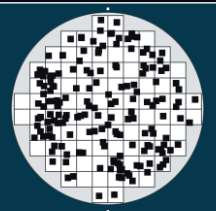
Deep Collaboration | Process Control Expertise | Portfolio

Process Steps



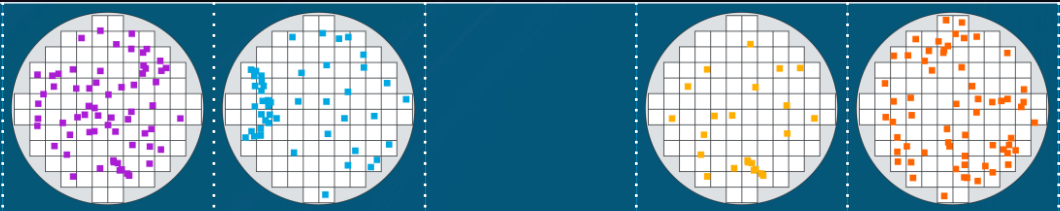
- One process loop shown
- Many process loops make up >1,000 process steps

1. Process Mapping



- Determine defectivity level in loop

2. Defect Source Analysis



- Determine the source of defectivity and variations

3. Process Control Optimization

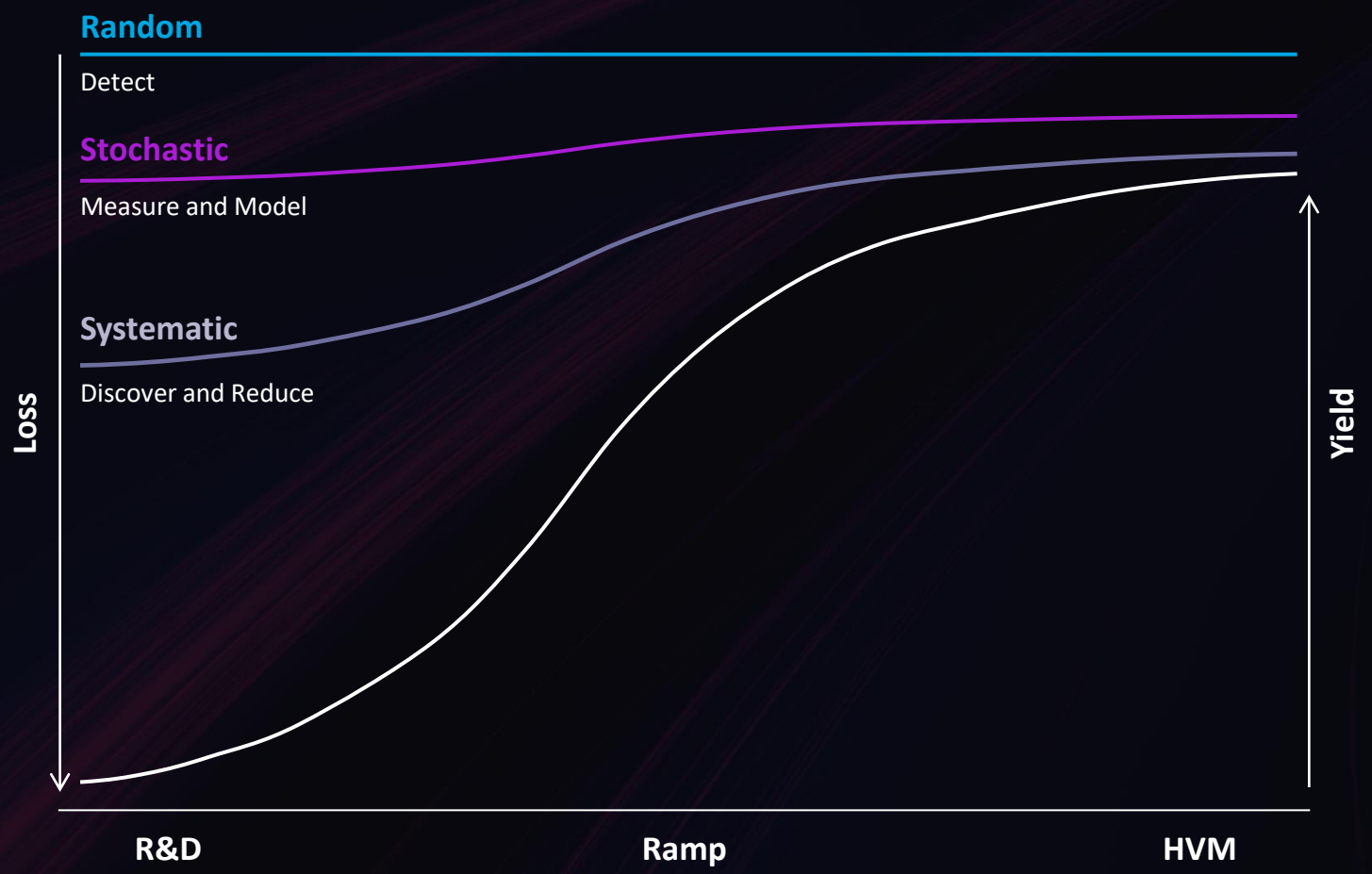


Chamber Monitoring

Inline Segmentation

Final Check

Process Control Improves the Productivity, Predictability and Profitability of a Fab Throughout the Manufacturing Lifecycle



Defect Types | Sources

Random: uncorrelated, non-predictable

- Process particles
- Equipment wear-out (bearings, seals, valves)
- Sporadic tool or material excursions

Stochastic: multi-parameter interactions, modellable

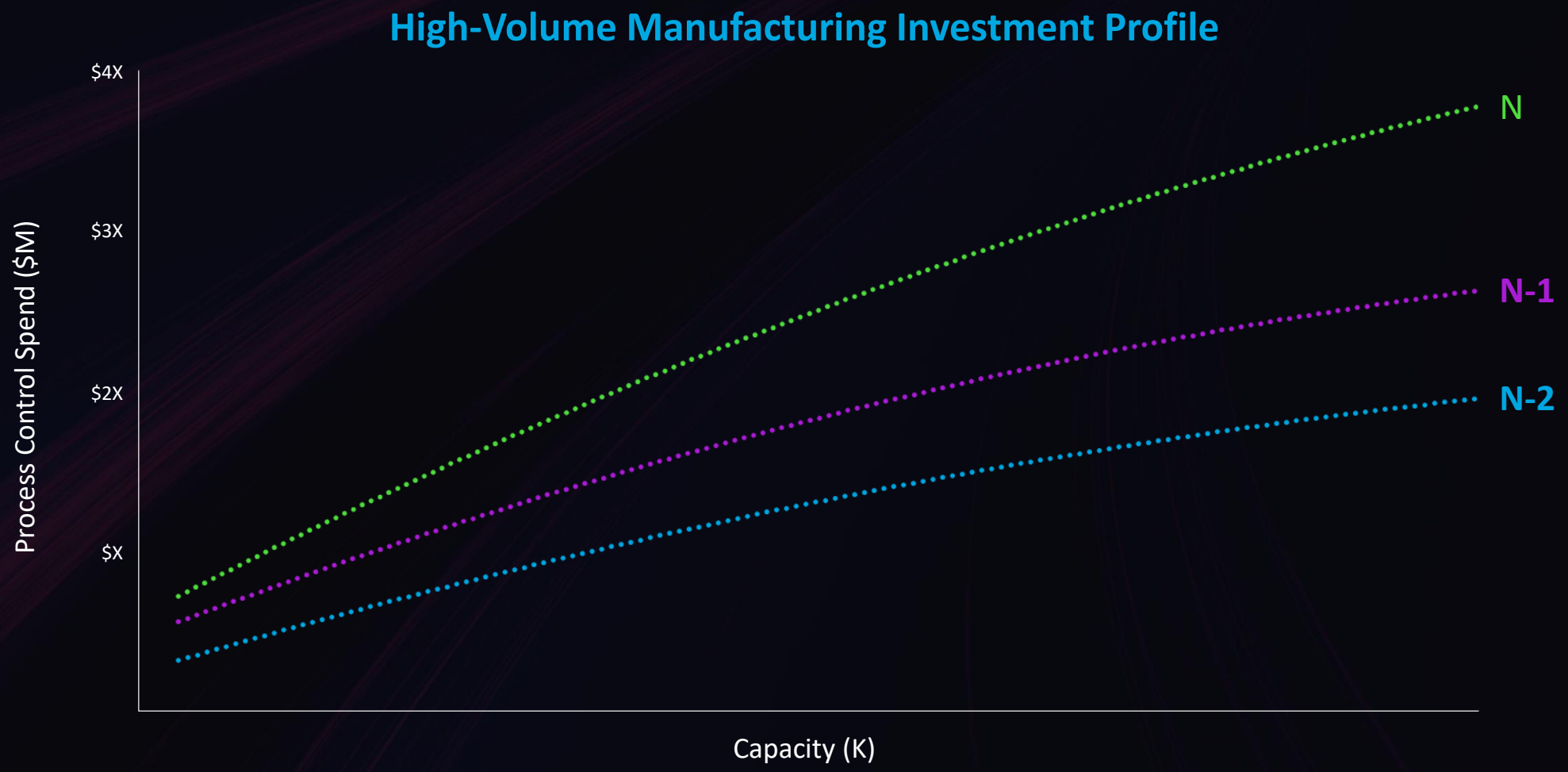
- Edge placement variability near design margins
- Pattern variability due to photon shot noise
- Etch bias and loading interacting with marginal CDs

Systematic: repeatable, spatially predictable

- Local layout effects (LLE/LDE)
- Etch, CMP and deposition non-uniformity signatures
- New product intro exposing latent margin weaknesses

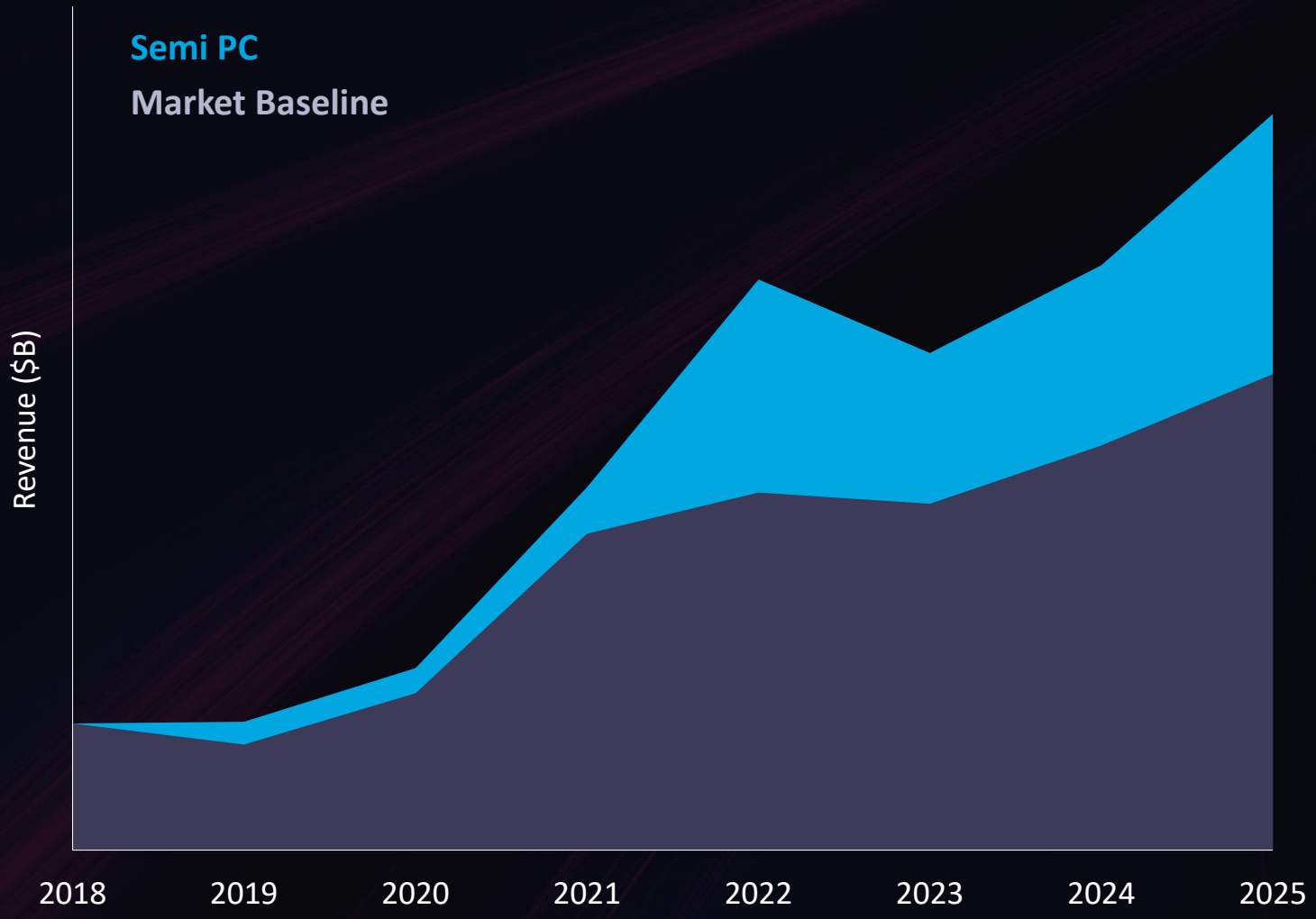
KLA's Products Deliver Solutions for High-Volume Manufacturing

Process Control Spend Grows with Capacity at Smaller Design Nodes



Source: KLA internal analysis

Enabling Customer Success Drives Our Success



Semi PC Relative Outperformance from 2018 – 2025

$8.5B

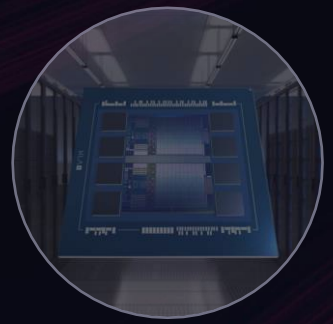
Process Control Intensity and Share Drivers

- Architectural complexity and scaling in logic, memory and advanced packaging
- Variability control in HVM

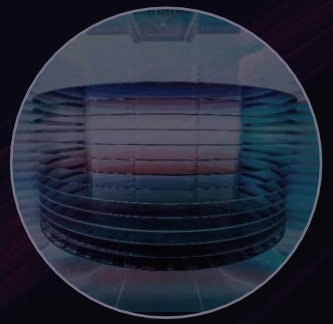
The KLA Operating Model Drives Semi PC Success



Customer Collaboration

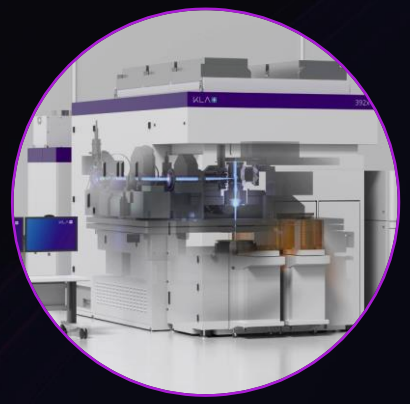


Technology Inflections



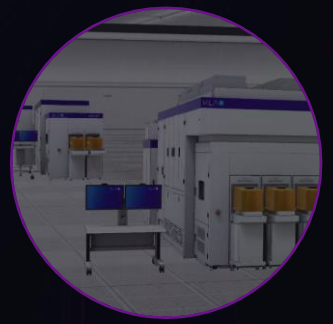
High-Volume Manufacturing

Product Innovation



Point Product Differentiation

Core technologies and AI produce world-class products



Co-Intelligent Solutions

Execution



2030 Target Model

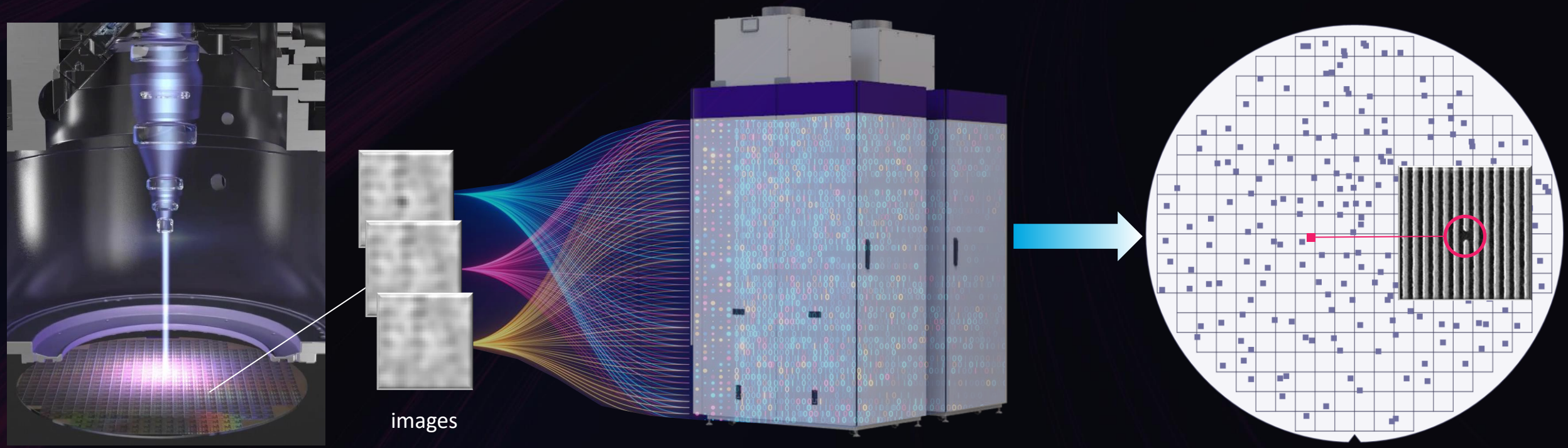
Our Products Require Best-in-Class Signal Coupled with Fast, Accurate Processing of Large Data Sets

Optical inspectors find critical defects with full wafer coverage

Collection
scan full wafer

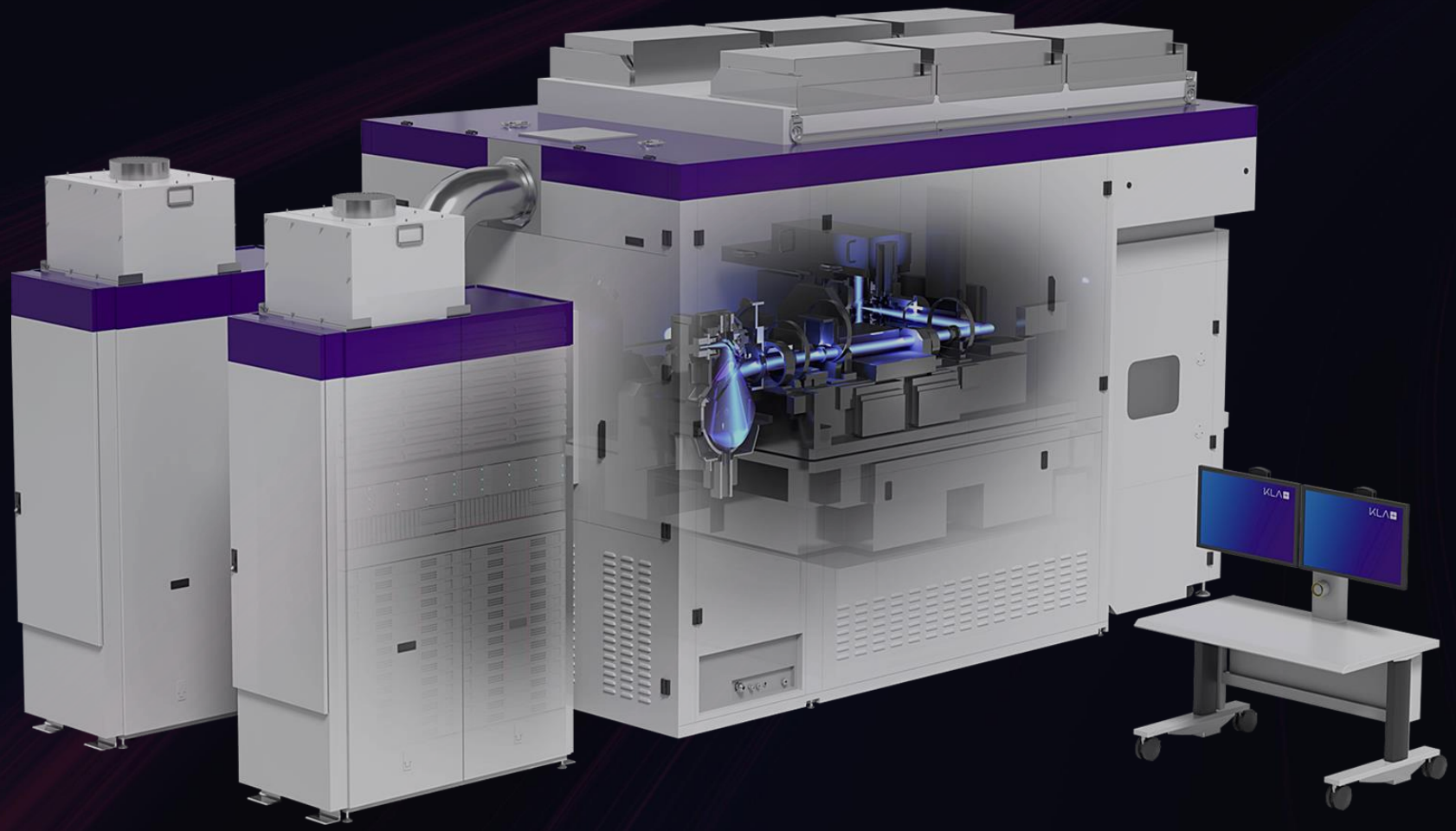
Processing
real time image analysis

Results
report defects



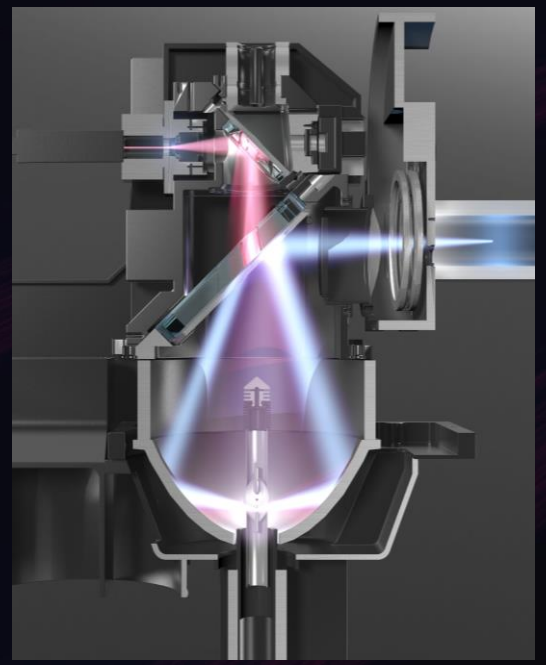
KLA's Core Technologies and Expertise

Optical Inspection Architecture

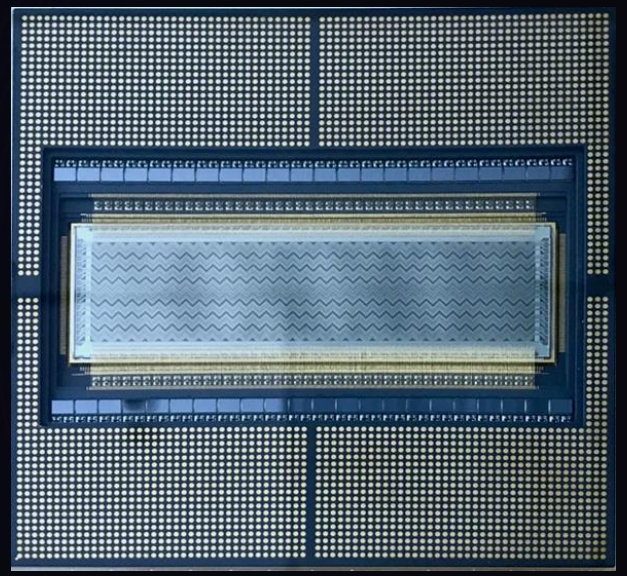


- light
- optics
- sensors
- pixels
- algorithms
- k factors

Gen5 BBP Optical Inspection Unique Technologies







Light

DUV broadband source that is >2x the brightness of the surface of the sun

Optics

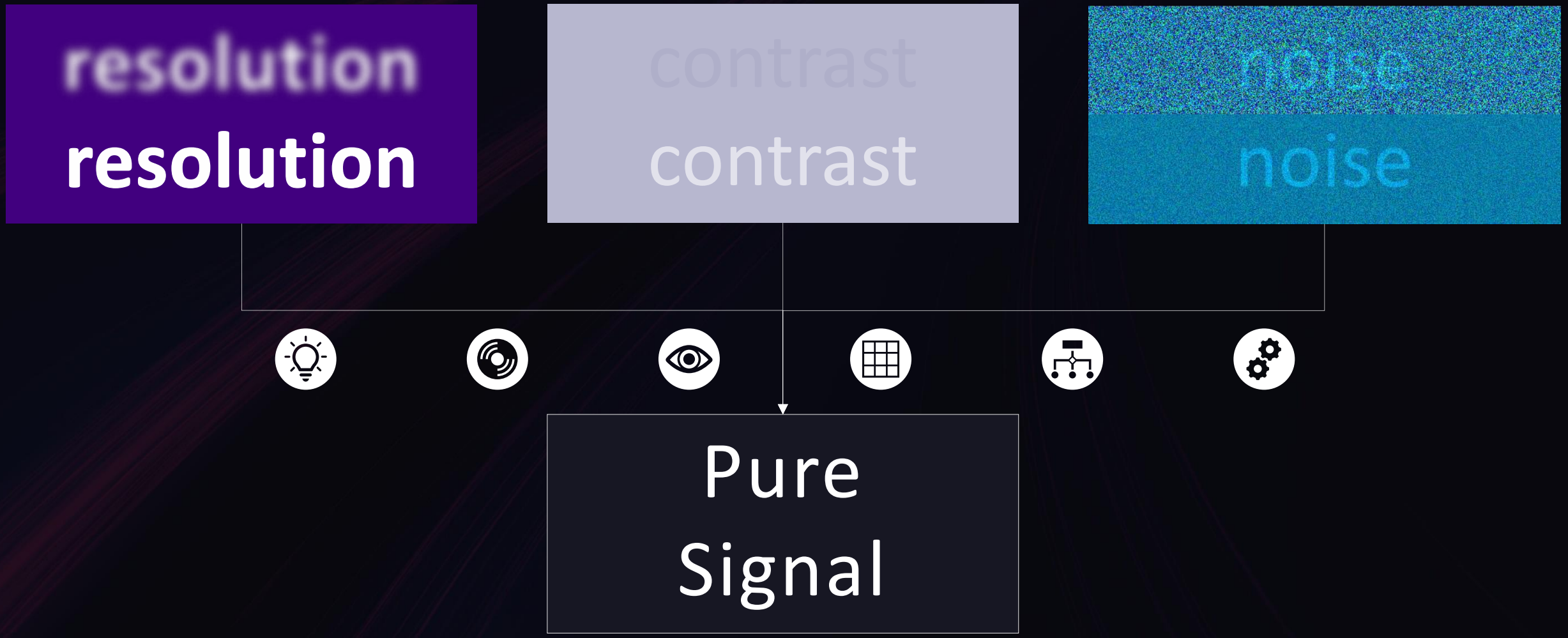
300kg catadioptric objective with CaF_2 crystal

Sensor

Massively parallel data collection equivalent to >2,000 smartphones simultaneously taking a photo **every second**

Best-in-Class Signal Required to Find Nanoscale Defects / Variations

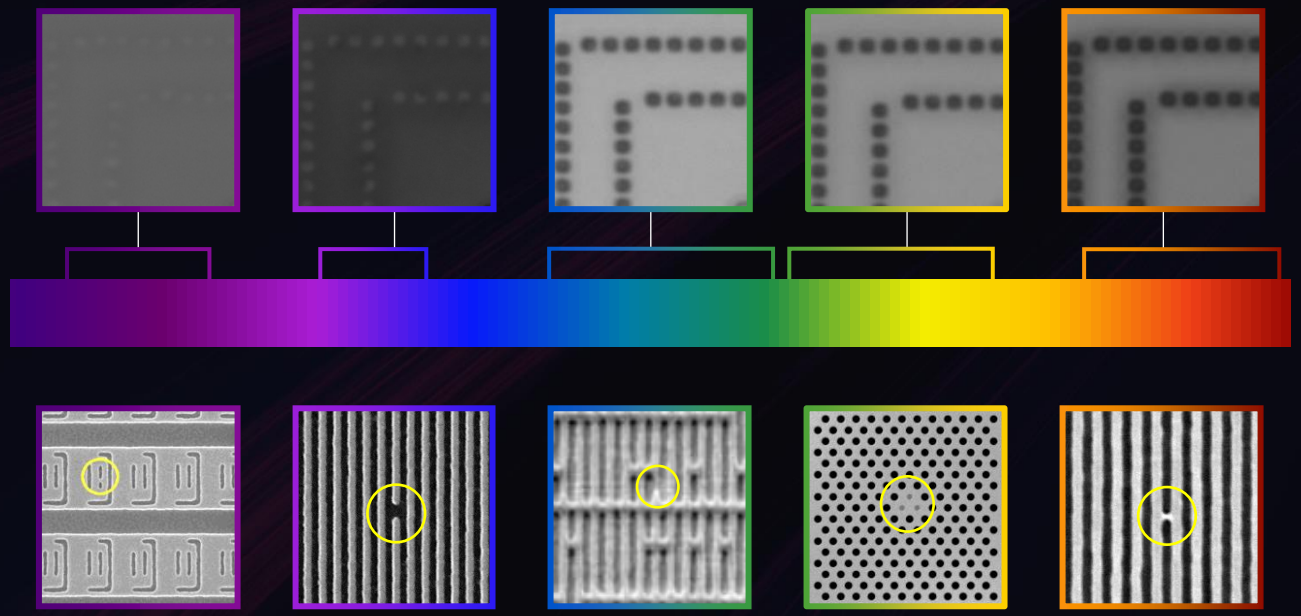
Advancements in Core Technologies Improve Resolution and Contrast, and Reduce Noise



Broadband Optical Inspection Delivers Unique Value

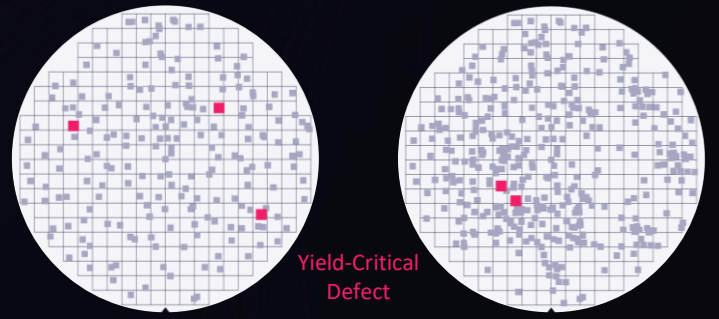
Sensitivity

Gen4 and Gen5 provide the widest wavelength range to find all defect types and sizes

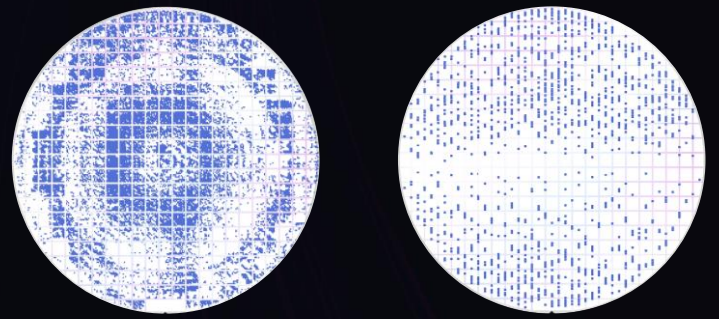


Coverage

Full wafer inspection is critically important as defects can occur anywhere



Yield-impacting defects can occur anywhere on the wafer

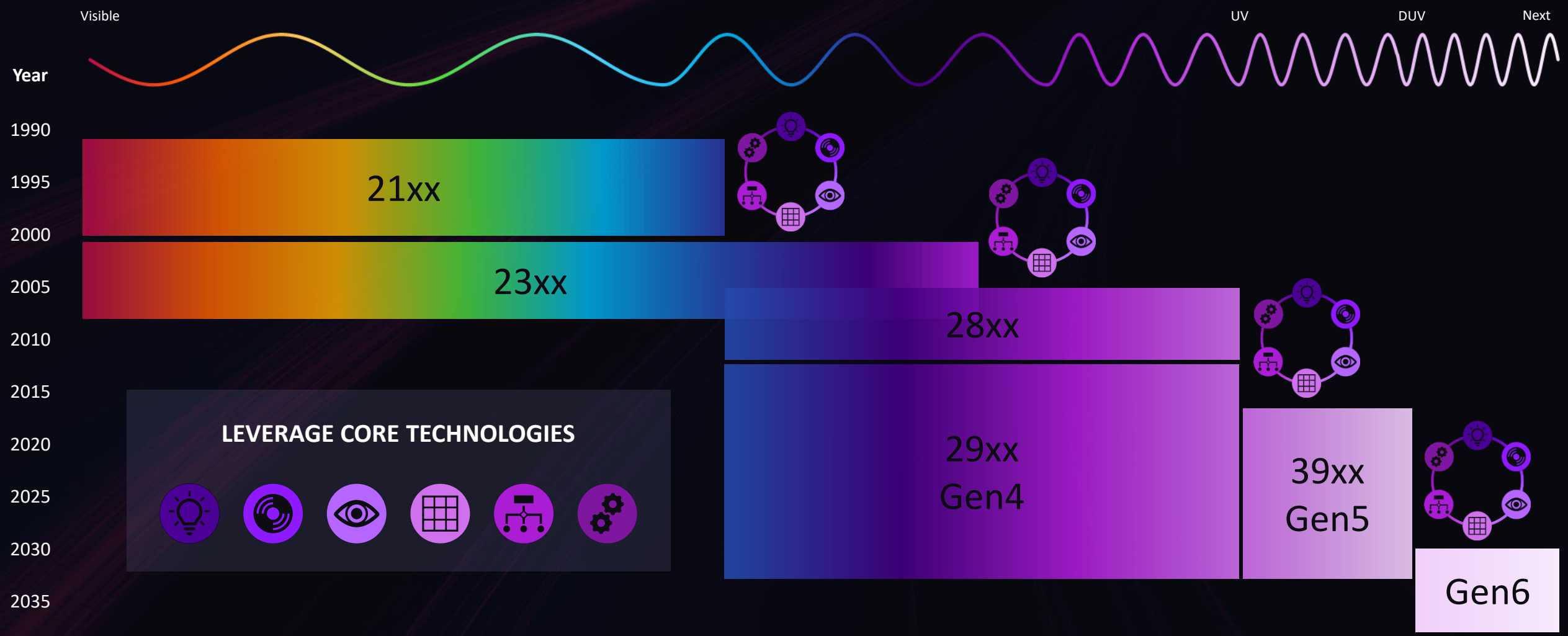


Wafer-level signatures help identify the source of defects

Broadband Platform Finds Critical Defects with Full Wafer Coverage Enabling Customer Success in R&D and HVM

Note: Defect image sources are SEMICON West 2021, SPIE 104510L, SPIE 1080909 and SPIE 115170U

Framework for Extending System Performance



Fast, Accurate Processing of Data Required for Wafer Coverage

An optical inspector collects

78 trillion pixels

contained in

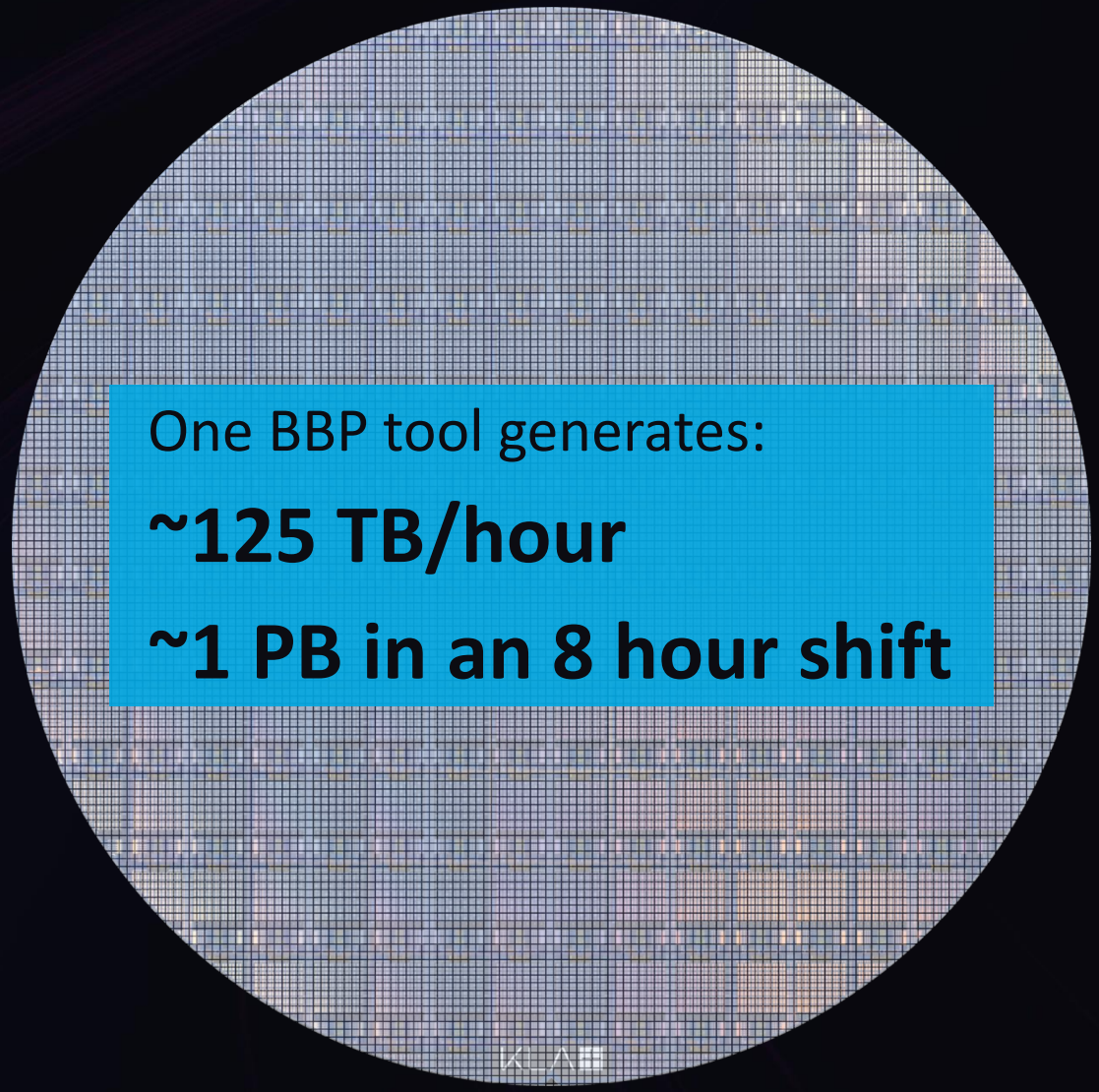
65 million frames per wafer

using

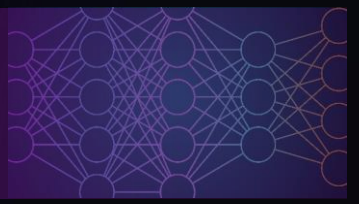
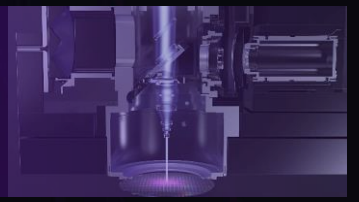
40 GPPS image sensors

and

60 GB/s GPU-based image computers running AI-based algorithms



AI at KLA: Five Layers Form a Differentiated Solution

5	**Applications**	Workflows for fab-wide process control	
4	**AI Models**	Models trained in optics, design and context	
3	**Software**	Infrastructure to manage high-volume, high data rate information in a production environment	
2	**High-Performance Compute**	GPU and CPU compute architecture that optimizes system performance and CoO	
1	**Physical Subsystems**	Light source, optics, sensor, etc. that produces pure signal	

KLA's AI Stack: Layer 2

High-Performance Compute with CPU/GPU Infrastructure Optimizes System Performance and Cost

Big Data Infrastructure

Improved Inspection Sensitivity is Achieved through Smaller Pixels, Enabling Detection of Smaller Defects

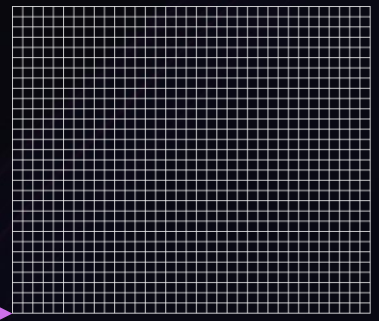
Large Pixel



BBP Optical Inspection Technologies that Enable Smaller Pixel:

- Advanced Optics / Objective
- Unique Broadband Light Source
- High-Performance Sensor
- Fine Precision Stage

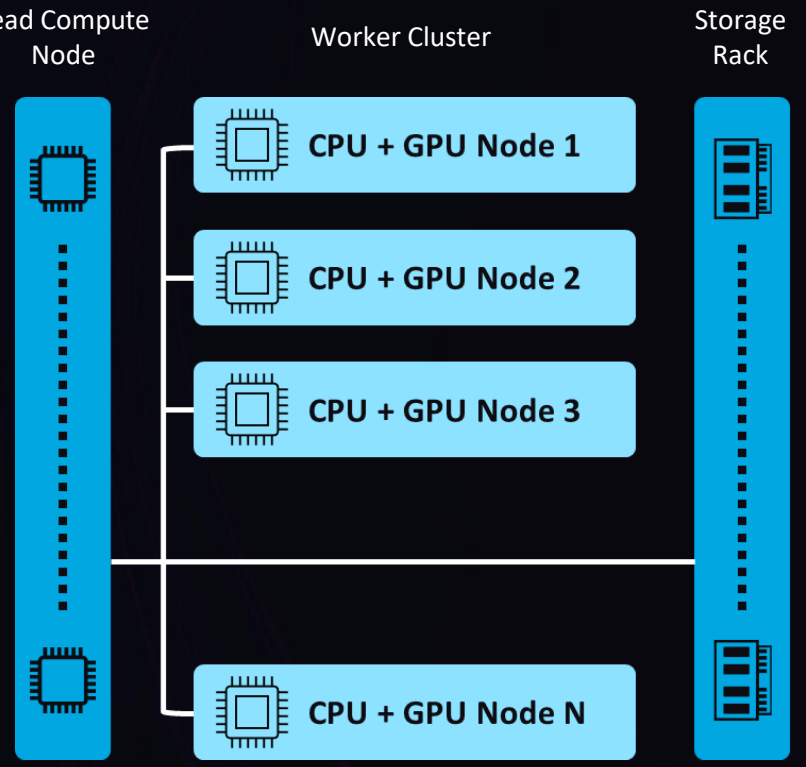
30nm Pixel



- Low cost of ownership system
- Under-sampled events

- ~78 trillion pixels per wafer
- Sub-pixel alignment
- **Requires high-performance, high data rate compute**

High-Performance Compute Architecture



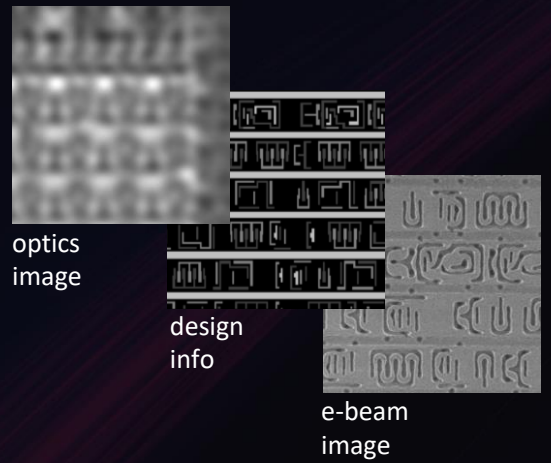
- GPU-based scalable compute
- Higher data rate

KLA's AI Stack: Layer 4

Models Trained in Optics, Design and Context

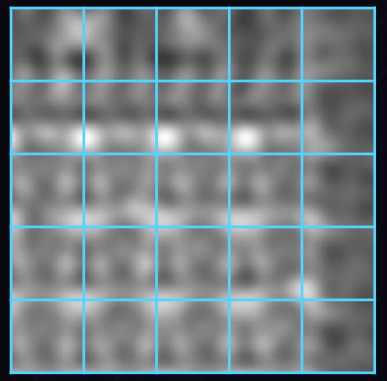
Optics | Design Info

- Sub-pixel aligned optics
- Design and e-beam images used as ground truth for defects of interest



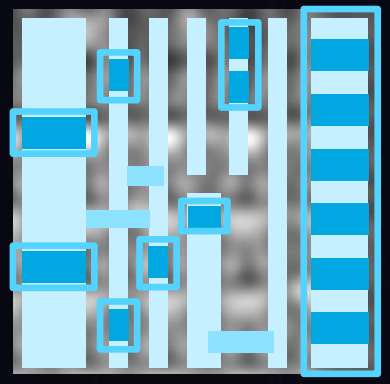
Pixel-Level Learning

- Learn features at every pixel



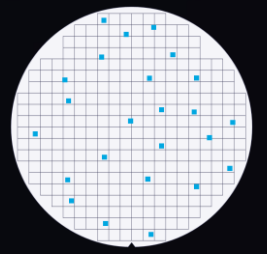
Context Mapping

- Learned segmentation from design



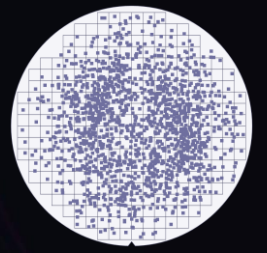
DL-Based Detection

- Enhanced defect capture and nuisance separation



With DL

- High capture of critical defects
- Nuisance suppressed



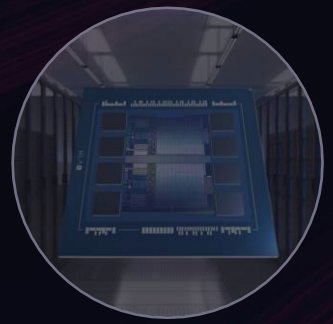
Without DL

- Low capture of critical defects
- Nuisance events dominate results

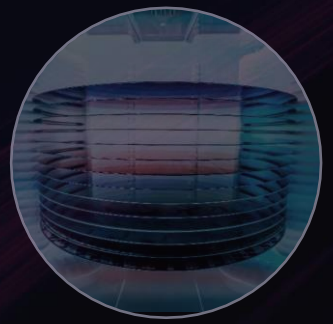
The KLA Operating Model Drives Semi PC Success



Customer Collaboration



Technology Inflections



High-Volume Manufacturing

Product Innovation



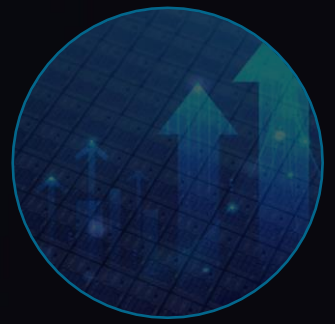
Point Product Differentiation



Co-Intelligent Solutions

Connected portfolio accelerates customer yield

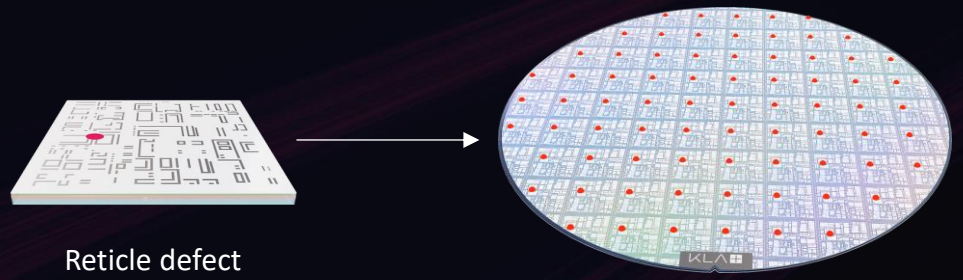
Execution



2030 Target Model

Reticle Quality is Critical; Inspection Challenges are Many

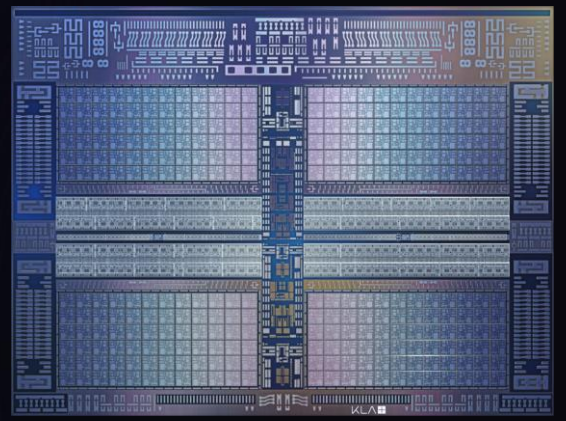
100% Capture of Defects is Required



Reticle defect

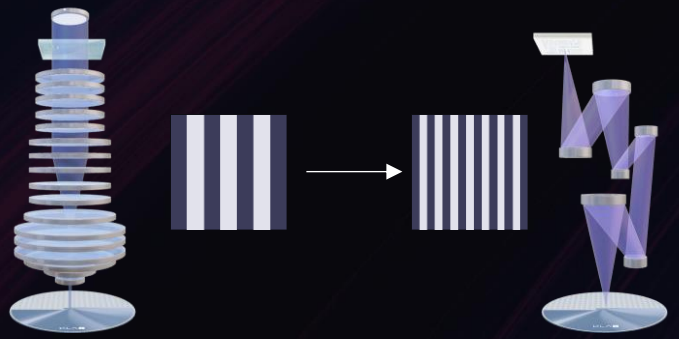
Repeater defect in every chip on the wafer: catastrophic to yield

Large Chip Size



Unique reference needed for inspection

EUV Prints Smaller Patterns, Smaller Defects

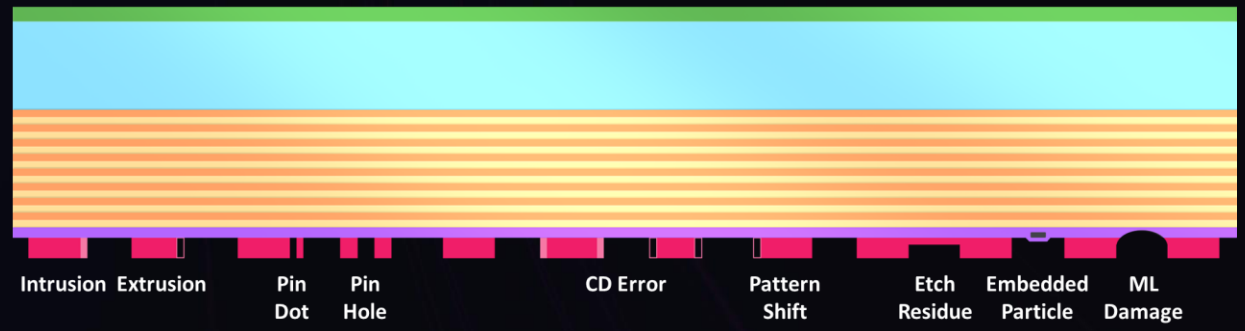


193nm Scanner

EUV Scanner

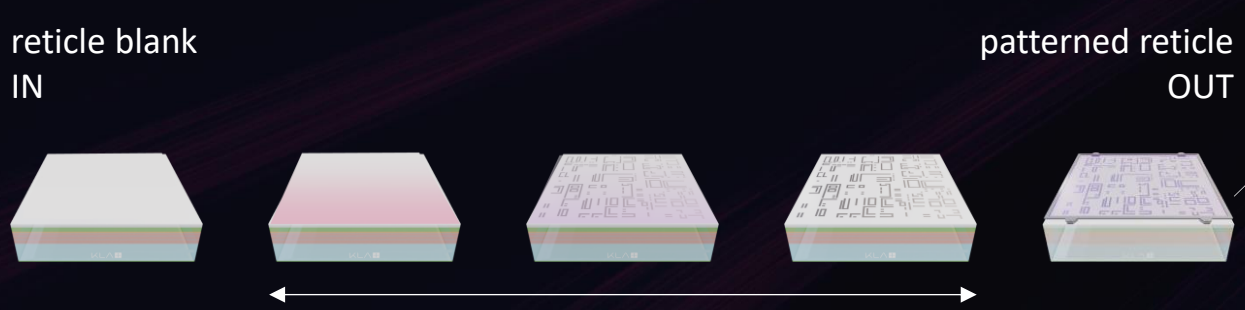
Improved inspection sensitivity required to detect smaller defects

Many Different Defect Types



Reticle Quality is Critical in Both the Mask Shop and IC Fab

Mask Shop – where reticles are made



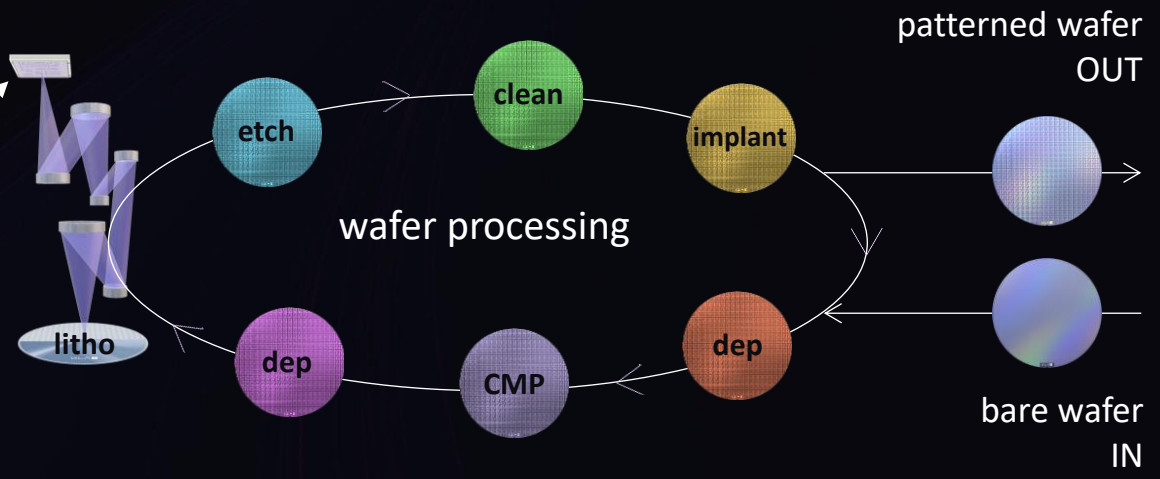
Defects

- Pattern and particles

Role of Reticle Inspection

- Reticle process qualification and monitoring
- Reticle outgoing qualification to reduce yield risk in IC fab

IC Fab – where reticles are used to make chips



Defects

- Particles that print

Role of Reticle Inspection | Print Check

- Reticle re-qual to reduce yield risk
- Reticle print verification for detection of repeater and stochastic defects

Number and Mix of Reticles Used in Chip Production Drive the Need for a Reticle Portfolio Inspection Strategy

Number of Reticles

=

Number of Designs

X

Number of Lithography Layers

X

Re-Spins

X

Extra Reticle for Cycle Time Improvement

Advanced Logic

Lithography Technology	193 | 193i **50+ Reticles**	EUV **20+ Reticles**
Inspection Technology	DUV **65+ Reticles**	High Sensitivity **3-5 Reticles**

Advanced Memory

Lithography Technology	193 | 193i **25+ Reticles**	EUV **6+ Reticles**
Inspection Technology	DUV **30+ Reticles**	High Sensitivity **1-2 Reticles**

*Excludes KrF, i-Line

KLA's Reticle Product Portfolio has Evolved to Support Reticle Inspection Challenges

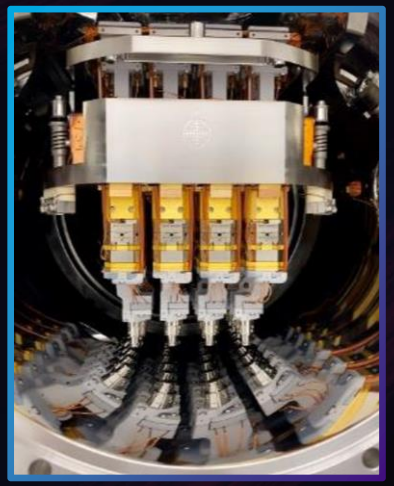
		Litho Wavelength	Use Case	Release Date
Mask Shop	Teron™ 610/630/640	Optical 193/193i	>70 nm	2009
	Teron™ 640e/eS/eS2	EUV	>3x nm	2017
	TeraBeam® 8xx	EUV	1x nm - 3x nm	2026
	Actinic 7xx	EUV	2x nm – 3x nm* *Requires NA Change	Early Demo Stage
IC Fab	Teron™ SL650/655/670	Optical 193/193i	Particles	2014
	Teron™ SL670e/XP/XP2	EUV	Particles	2018
	Gen5 Agera™ Wafer Print Check	EUV	Particles that Print	2019

TeraBeam® 8xx: High Resolution Reticle Pattern Inspection

Critical for Finding the Smallest Defects Written on Reticle

TeraBeam® 8xx











Complex, Differentiated, Multi-Column e-Beam Technology Captures Critical Defects on Advanced Reticles

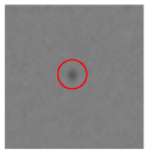
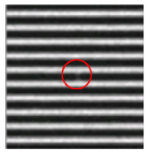
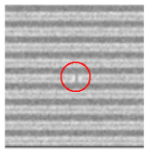
TeraBeam® 8xx: High Resolution Reticle Pattern Inspection

Introduction – Benchmark of EUV Pattern Inspection Tools

Unleash Innovation

KLA

- e-Beam tool can be the solution to fulfill the resolution needs of advance nodes.

	DUV	EUV	e-Beam
Source	193 nm-laser	13.5 nm-plasma	e-gun (<0.1 nm)
Image optics	Flexible illumination & refractive optics	Off-axis illumination & reflective optics	Electric & magnetic optics
Pattern resolution (pitch 1X, nm)	36 nm	24 nm	20 nm
Inspection time	1x	1.5x	Target <2x
 Line/Space Pattern (with programed defect)			

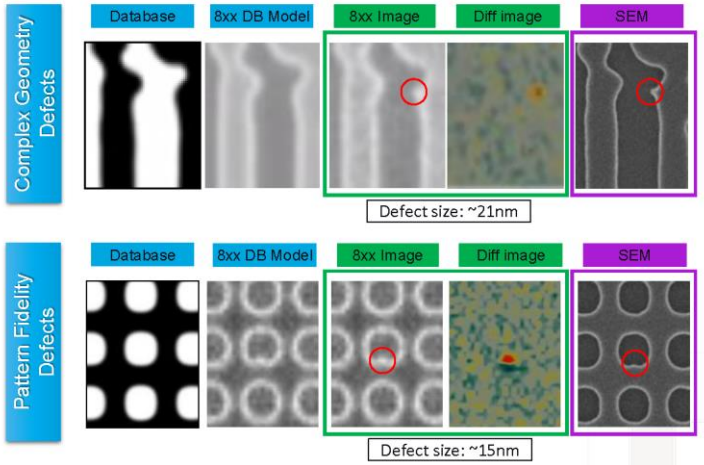
© 2026 TSMC, Ltd TSMC Property

Sensitivity Benefits from KLA 8xx e-Beam Inspection

Unleash Innovation

KLA

- High resolution of e-Beam inspection tool to meet fidelity and sensitivity demands



KLA 8xx offering superior defect detection capability on complex OPC patterns.

© 2026 TSMC, Ltd TSMC Property

e-Beam: Ultimate Resolution for the Smallest Critical Defects

Source: "High-resolution multi-column electron beam inspection for EUV reticle development and production," SPIE Advanced Lithography + Patterning, Paper 13981-231, 2026.

High-Volume Manufacturing Reticle Requalification Requires Co-Intelligent Solutions

No Escape Risk at Optimized Cost



Defense 1

1 inspection for every ~100-200 wafers printed

Teron™ SL670e XP2

high sensitivity direct reticle inspection



Defense 2

1 inspection for every ~200-400 wafers printed

Gen5 (39xx) Agera™

high sensitivity wafer inspection, repeater defects

Defense 3

1 inspection for every ~2,000-5,000 wafers printed

Actinic

through pellicle reticle inspection

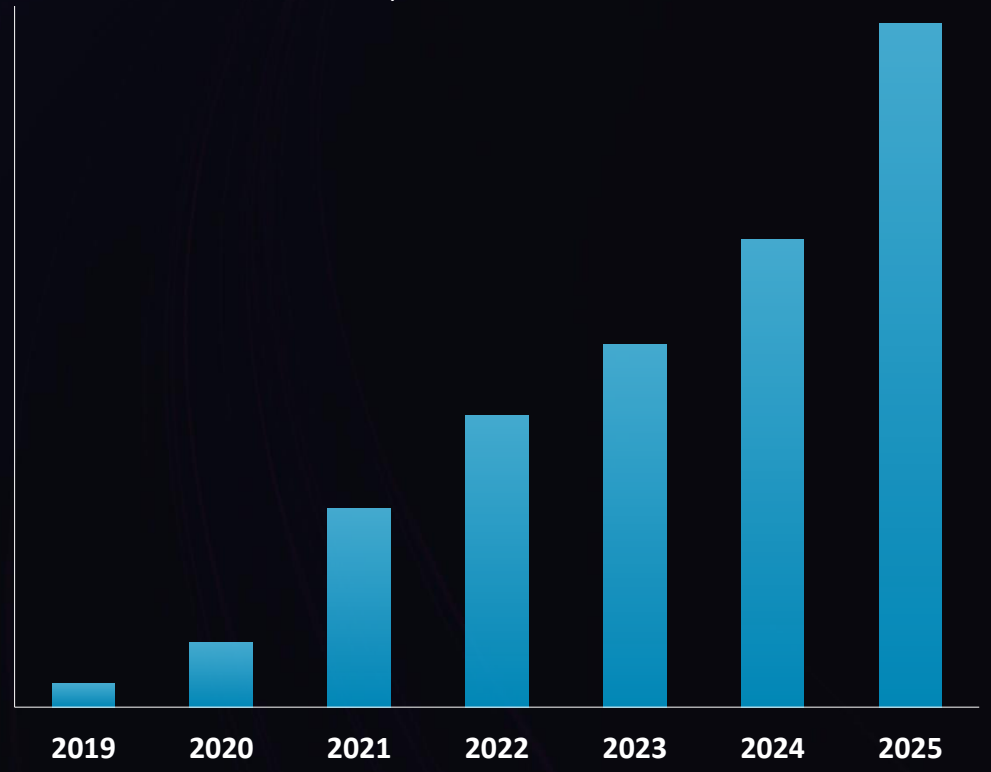
Inspection Frequency

Cost Per Inspection

Significant Adoption Across Logic and DRAM

Cumulative Units for Reticle Requal in IC Fab

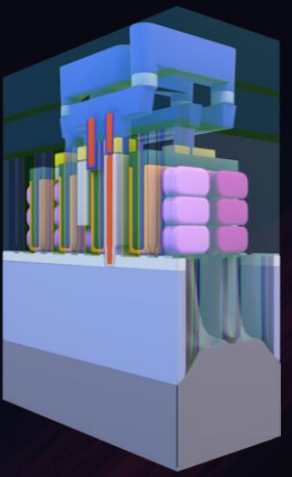
DUV Reticle Requal and Gen5 Wafer Print Check



KLA's e-Beam Portfolio: Point Product Differentiation Supports Co-Intelligent Solutions

1 If you can detect with optical, you always use optical

1nm Range of Defect Sizes Covered by High Sensitivity Inspection **50nm**



2 GAA Transistor Architecture

- New subset of defect types (buried and electrical) requires co-intelligence
- Gen4 BBP optical with longer wavelengths and e-beam sensitivity

3 Co-Intelligent Solution: Optical and e-Beam

- BBP optical inspection leverages e-beam information to accelerate setup, enabling critical defect detection in high-volume manufacturing

KLA's e-Beam Portfolio: Point Product Differentiation Supports Co-Intelligent Solutions



eSi50™
single beam inspection



eSV100™
multi-beam inspection



eDRX1™
review



eM200™
metrology

Discover Critical Defects

Identify and Classify Defects

Measure Critical Parameters

Differentiated e-Beam Portfolio Supports Industry Inflections: ~$400M Business in CY25

BBP and e-Beam Co-Intelligent Solution: Fast Cycles | Higher Yield

eSi50™ single beam



e-beam inspection baseline



eSV100™ multi-beam

Gen4 | Gen5 BBP optical inspection



wafer-level monitoring

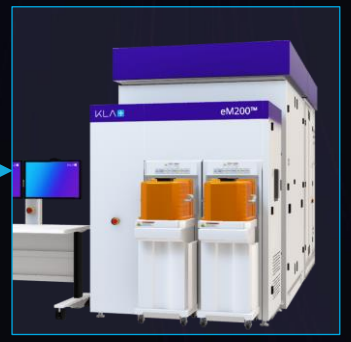


AI Compute

eDRX1™ review

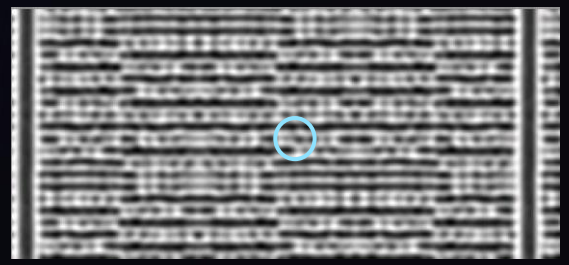


e-beam validation

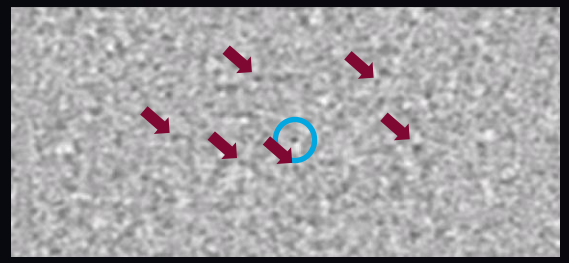


eM200™ metrology

BBP Test Image



BBP Difference Image



AI and e-Beam Synergy Produce Actionable Optical Insight

BBP Optical Inspection Success

~$8.5B

BBP Optical Inspection Revenue
CY22 – CY25

~30%

BBP Optical Inspection CAGR
CY19 – CY25

>600

Gen4 Systems Shipped
release to end of CY25

>180

Gen5 Systems Shipped
release to end of CY25

Gen4
R90x | 29xx





Gen5
39xx

BBP Optical Inspection is Required for Yield, Reliability and Scale in Advanced Chip Manufacturing

2030 Systems Target Model



Strategy Drives Outperformance



Strategy: Provide leading technologies, differentiated solutions and superior customer experiences by leveraging our operating model

Collaborate

Logic Inflections

Inflections in logic architectures result in ~$1B incremental revenue for KLA over two design nodes

Memory Demand

Challenges in DDR and HBM result in $450M incremental intensity gain for KLA in CY25

Advanced Packaging

Significant increase in adoption of KLA's process control solutions drives ~$950M revenue in CY25

High-Volume Manufacturing

Process control spend is increasing as process nodes shrink

Innovate

BBP Optical Inspection

Continued strength in demand for optical inspection for the leading-edge results in ~$8.5B revenue

CY22 – CY25

Reticle Inspection

KLA's diversified technology portfolio supports all critical reticle inspection requirements in mask shop and IC fab

e-Beam Evolution

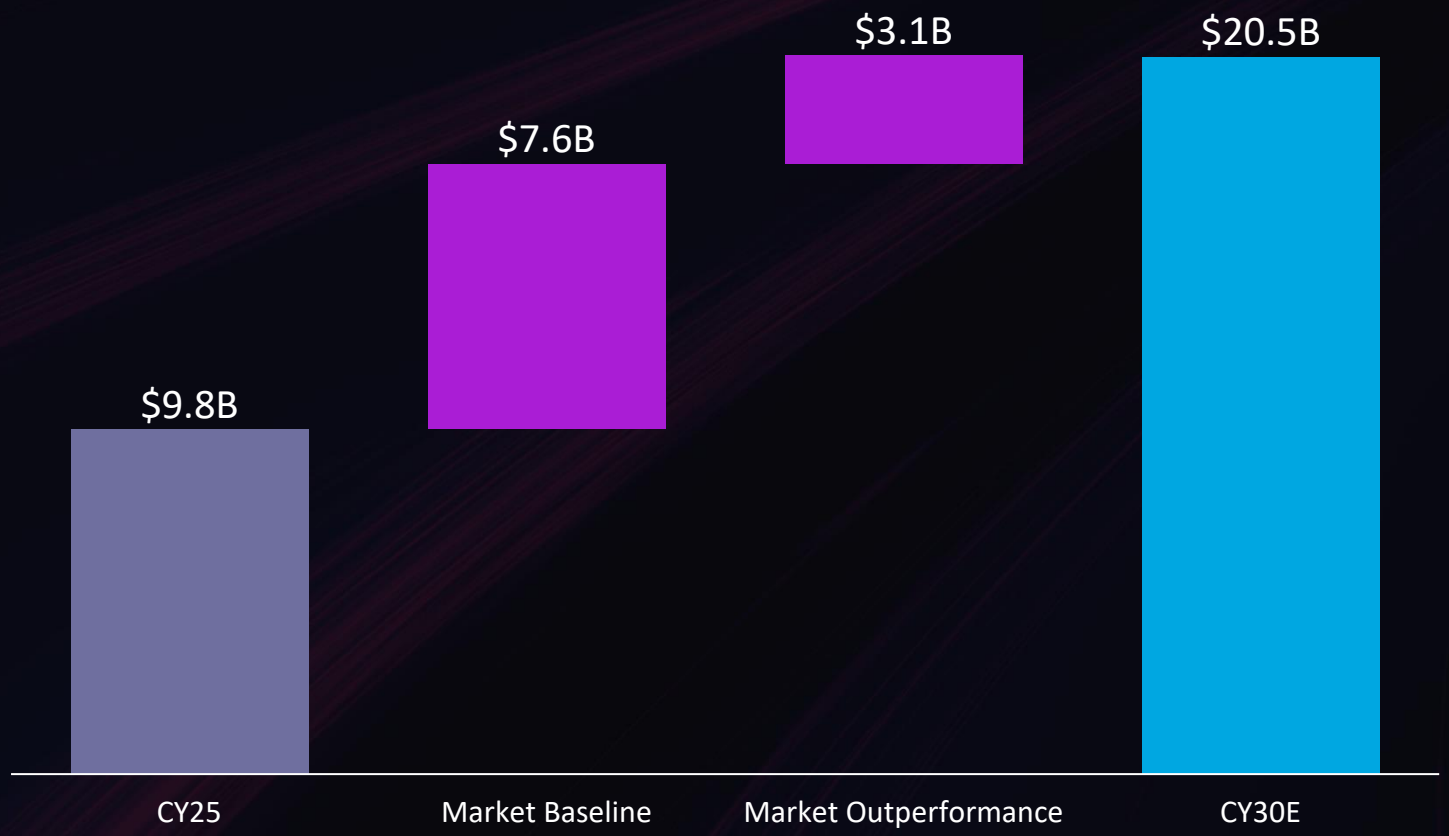
e-Beam portfolio supports strategy of deploying co-intelligent solutions to support leading-edge chip technologies

Portfolio Expansion

- X-Ray Metrology
- Chemical Analysis
- Data Analytics
- Plasma Dicing
- Component Inspection
- PCB Inspection
- …

Execute

KLA Systems 2030 Target Model



11% - 13% CAGR
Market

15% - 17% CAGR
Systems Revenue

Semi PC CAGR ~4.5%
> Market CAGR

Achieving Our CY30 Targets through Continued Product Differentiation and Portfolio Solutions for R&D and HVM